Financial Risk Analysis

Project Report

Project Details:

Project Name	Financial Risk Analysis Project
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Date	15-02-2025

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1. Project Background

In the ever-evolving landscape of modern finance, businesses face the continuous challenge of effectively managing their debt obligations to uphold a strong credit standing and support sustainable growth. Investors and stakeholders are increasingly focused on identifying companies that can navigate complex financial dynamics while ensuring operational efficiency, profitability, and long-term stability.

A critical tool in assessing a company's financial health is the balance sheet, which provides a snapshot of assets, liabilities, and shareholder equity, offering insights into its overall financial performance. However, due to the vast volume of historical financial data and the intricacies involved in evaluating these metrics, it becomes difficult for businesses and investors to make timely, data-driven decisions regarding the risk of defaults and financial viability.

The problem lies in the lack of an automated, data-driven mechanism to effectively analyse financial statements, particularly for predicting potential financial distress or defaults. There is a need for a Financial Health Assessment Tool powered by machine learning techniques to analyse historical financial data, identify key trends, and assess the likelihood of companies meeting their financial obligations in the future. This tool would empower businesses and investors with accurate, predictive insights to improve debt management, evaluate credit risk exposure, and make informed investment decisions.

2. Objective

The objective of this project is to develop a predictive model using machine learning techniques to assess the likelihood of a company defaulting on its financial obligations in terms of net worth in the upcoming year. By leveraging historical financial data and key financial metrics (such as debt management practices, liquidity ratios, and debt-to-equity ratios), the model will:

- **Analyse Debt Management**: Identify patterns and trends in how companies manage their debts, evaluate their ability to meet financial obligations efficiently, and detect signs of potential default risks.
- **Evaluate Credit Risk**: Assess the exposure to credit risk by analysing critical financial indicators and ratios, predicting the likelihood of default, and informing investors' decisions regarding the financial stability of companies.
- **Enable Informed Decision-Making**: Provide businesses and investors with actionable insights into the financial health and creditworthiness of companies, helping them make better, data-driven investment decisions.
- **Facilitate Proactive Risk Mitigation**: Offer early warning signals about potential financial distress, allowing the organization to implement proactive strategies to mitigate risk, manage debt more effectively, and ensure long-term financial stability.

By achieving these objectives, the tool will empower businesses and investors with a robust mechanism to navigate financial complexities and make informed, strategic decisions.

3. Data description

The data consists of financial metrics from the balance sheets of different companies. The detailed data dictionary is given below.

- Networth Next Year: Net worth of the customer in the next year
- Total assets: Total assets of customer
- Net worth: Net worth of the customer of the present year
- Total income: Total income of the customer
- Change in stock: Difference between the current value of the stock and the value of stock in the last trading day
- Total expenses: Total expenses done by the customer
- Profit after tax: Profit after tax deduction
- PBDITA: Profit before depreciation, income tax, and amortization
- PBT: Profit before tax deduction
- Cash profit: Total Cash profit
- PBDITA as % of total income: PBDITA / Total income
- PBT as % of total income: PBT / Total income
- PAT as % of total income: PAT / Total income
- Cash profit as % of total income: Cash Profit / Total income
- PAT as % of net worth: PAT / Net worth
- Sales: Sales done by the customer
- Income from financial services: Income from financial services
- Other income: Income from other sources
- Total capital: Total capital of the customer
- Reserves and funds: Total reserves and funds of the customer
- Borrowings: Total amount borrowed by the customer
- Current liabilities & provisions: current liabilities of the customer
- Deferred tax liability: Future income tax customer will pay because of the current transaction
- Shareholders funds: Amount of equity in a company which belongs to shareholders
- Cumulative retained profits: Total cumulative profit retained by customer
- Capital employed: Current asset minus current liabilities
- TOL/TNW: Total liabilities of the customer divided by Total net worth

- Total term liabilities / tangible net worth: Short + long term liabilities divided by tangible net worth
- Contingent liabilities / Net worth (%): Contingent liabilities / Net worth
- Contingent liabilities: Liabilities because of uncertain events
- Net fixed assets: The purchase price of all fixed assets
- Investments: Total invested amount
- Current assets: Assets that are expected to be converted to cash within a year
- Net working capital: Difference between the current liabilities and current assets
- Quick ratio (times): Total cash divided by current liabilities
- Current ratio (times): Current assets divided by current liabilities
- Debt to equity ratio (times): Total liabilities divided by its shareholder equity
- Cash to current liabilities (times): Total liquid cash divided by current liabilities
- Cash to average cost of sales per day: Total cash divided by the average cost of the sales
- Creditors turnover: Net credit purchase divided by average trade creditors
- Debtors turnover: Net credit sales divided by average accounts receivable
- Finished goods turnover: Annual sales divided by average inventory
- WIP turnover: The cost of goods sold for a period divided by the average inventory for that period
- Raw material turnover: Cost of goods sold is divided by the average inventory for the same period
- Shares outstanding: Number of issued shares minus the number of shares held in the company
- Equity face value: cost of the equity at the time of issuing
- EPS: Net income divided by the total number of outstanding shares
- Adjusted EPS: Adjusted net earnings divided by the weighted average number of common shares outstanding on a diluted basis during the plan year
- Total liabilities: Sum of all types of liabilities
- PE on BSE: Company's current stock price divided by its earnings per share

Note: A company will not be tagged as a defaulter if its net worth next year is positive, or else, it'll be tagged as a defaulter.

4. Exploratory Data Analysis

4.1. Data structure

After importing NumPy and Pandas libraries for tasks like data reading, data computation and data manipulation and loading data, the following observations regarding the data structure were noted.

	Num	Networth_Next_Year	Total_assets	Net_worth	Total_income	Change_in_stock	Total_expenses	Profit_after_tax	PBDITA	PBT	...	Debtors_turnover
0	1	395.3	827.6	336.5	534.1	13.5	508.7	38.9	124.4	64.6	...	5.65
1	2	36.2	67.7	24.3	137.9	-3.7	131.0	3.2	5.5	1.0	...	NaN
2	3	84.0	238.4	78.9	331.2	-18.1	309.2	3.9	25.8	10.5	...	2.51
3	4	2041.4	6883.5	1443.3	8448.5	212.2	8482.4	178.3	418.4	185.1	...	1.91
4	5	41.8	90.9	47.0	388.6	3.4	392.7	-0.7	7.2	-0.6	...	68.00

Debtors_turnover	Finished_goods_turnover	WIP_turnover	Raw_material_turnover	Shares_outstanding	Equity_face_value	EPS	Adjusted_EPS	Total_liabilities	PE_on_BSE
5.65	3.99	3.37	14.87	8760056.0	10.0	4.44	4.44	827.6	NaN
NaN	NaN	NaN	NaN	NaN	NaN	0.00	0.00	67.7	NaN
2.51	17.67	8.76	8.35	NaN	NaN	0.00	0.00	238.4	NaN
1.91	18.14	18.62	11.11	10000000.0	10.0	17.60	17.60	6883.5	NaN
68.00	45.87	28.67	19.93	107315.0	100.0	-6.52	-6.52	90.9	NaN

Table 1. Data structure

OBSERVATION	VARIABLES
4256	51

4.2. Data types

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4256 entries, 0 to 4255
Data columns (total 51 columns):
 #   Column                                     Non-Null Count  Dtype  
---  ------                                     --------------  -----  
 0   Num                                        4256 non-null   int64  
 1   Networth_Next_Year                         4256 non-null   float64
 2   Total_assets                               4256 non-null   float64
 3   Net_worth                                  4256 non-null   float64
 4   Total_income                               4025 non-null   float64
 5   Change_in_stock                            3706 non-null   float64
 6   Total_expenses                             4091 non-null   float64
 7   Profit_after_tax                           4102 non-null   float64
 8   PBDITA                                     4102 non-null   float64
 9   PBT                                        4102 non-null   float64
 10  Cash_profit                                4102 non-null   float64
 11  PBDITA_as_perc_of_total_income             4177 non-null   float64
 12  PBT_as_perc_of_total_income                4177 non-null   float64
 13  PAT_as_perc_of_total_income                4177 non-null   float64
 14  Cash_profit_as_perc_of_total_income        4177 non-null   float64
 15  PAT_as_perc_of_net_worth                   4256 non-null   float64
 16  Sales                                      3951 non-null   float64
 17  Income_from_fincial_services               3145 non-null   float64
 18  Other_income                               2700 non-null   float64
 19  Total_capital                              4251 non-null   float64
 20  Reserves_and_funds                         4158 non-null   float64
 21  Borrowings                                 3825 non-null   float64
 22  Current_liabilities_&_provisions           4146 non-null   float64
 23  Deferred_tax_liability                     2887 non-null   float64
 24  Shareholders_funds                         4256 non-null   float64
 25  Cumulative_retained_profits                4211 non-null   float64
 26  Capital_employed                           4256 non-null   float64
 27  TOL_to_TNW                                 4256 non-null   float64
 28  Total_term_liabilities__to__tangible_net_worth  4256 non-null   float64
 29  Contingent_liabilities__to__Net_worth_perc  4256 non-null   float64
 30  Contingent_liabilities                     2854 non-null   float64
 31  Net_fixed_assets                           4124 non-null   float64
 32  Investments                                2541 non-null   float64
 33  Current_assets                             4176 non-null   float64
 34  Net_working_capital                        4219 non-null   float64
 35  Quick_ratio_times                          4151 non-null   float64
 36  Current_ratio_times                        4151 non-null   float64
 37  Debt_to_equity_ratio_times                 4256 non-null   float64
 38  Cash_to_current_liabilities_times          4151 non-null   float64
 39  Cash_to_average_cost_of_sales_per_day      4156 non-null   float64
 40  Creditors_turnover                         3865 non-null   float64
 41  Debtors_turnover                           3871 non-null   float64
 42  Finished_goods_turnover                    3382 non-null   float64
 43  WIP_turnover                               3492 non-null   float64
 44  Raw_material_turnover                      3828 non-null   float64
 45  Shares_outstanding                         3446 non-null   float64
 46  Equity_face_value                          3446 non-null   float64
 47  EPS                                        4256 non-null   float64
 48  Adjusted_EPS                               4256 non-null   float64
 49  Total_liabilities                          4256 non-null   float64
 50  PE_on_BSE                                  1629 non-null   float64
dtypes: float64(50), int64(1)
memory usage: 1.7 MB
```

Table 2. Data types

- The data set contains 51 variables indexed from 0-50.
- All the columns are numerical columns, where 50 columns are of float data type and 1 column is integer data type.
- The dataset contains several missing values.
- These variables are essential for evaluating a company's financial health, debt management, creditworthiness and profitability.

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4.3. Statistical Summary

	count	mean	std	min	25%	50%	75%	max
Num	4256.00	2128.50	1228.75	1.00	1064.75	2128.50	3192.25	4256.00
Networth_Next_Year	4256.00	1344.74	15936.74	-74265.60	3.98	72.10	330.82	805773.40
Total_assets	4256.00	3573.62	30074.44	0.10	91.30	315.50	1120.80	1176509.20
Net_worth	4256.00	1351.95	12961.31	0.00	31.48	104.80	389.85	613151.60
Total_income	4025.00	4688.19	53918.95	0.00	107.10	455.10	1485.00	2442828.20
Change_in_stock	3706.00	43.70	436.92	-3029.40	-1.80	1.60	18.40	14185.50
Total_expenses	4091.00	4356.30	51398.09	-0.10	96.80	426.80	1395.70	2366035.30
Profit_after_tax	4102.00	295.05	3079.90	-3908.30	0.50	9.00	53.30	119439.10
PBDITA	4102.00	605.94	5646.23	-440.70	6.93	36.90	158.70	208576.50
PBT	4102.00	410.26	4217.42	-3894.80	0.80	12.60	74.17	145292.60
Cash_profit	4102.00	408.27	4143.93	-2245.70	2.90	19.40	96.25	176911.80
PBDITA_as_perc_of_total_income	4177.00	3.18	172.26	-6400.00	4.97	9.68	16.47	100.00
PBT_as_perc_of_total_income	4177.00	-18.20	419.91	-21340.00	0.56	3.34	8.94	100.00
PAT_as_perc_of_total_income	4177.00	-20.03	423.58	-21340.00	0.35	2.37	6.42	150.00
Cash_profit_as_perc_of_total_income	4177.00	-9.02	299.96	-15020.00	2.00	5.66	10.73	100.00
PAT_as_perc_of_net_worth	4256.00	10.17	61.53	-748.72	0.00	8.04	20.20	2466.67
Sales	3951.00	4645.68	53080.90	0.10	113.35	468.60	1481.20	2384984.40
Income_from_fincial_services	3145.00	81.36	1042.76	0.00	0.50	1.90	9.80	51938.20
Other_income	2700.00	55.95	1178.42	0.00	0.40	1.50	6.20	42856.70
Total_capital	4251.00	224.56	1684.95	0.10	13.20	42.60	103.15	78273.20
Reserves_and_funds	4158.00	1210.56	12816.23	-6525.90	5.30	55.15	282.52	625137.80
Borrowings	3825.00	1176.25	8581.25	0.10	24.40	99.80	358.30	278257.30
Current_liabilities_&_provisions	4146.00	960.63	9140.54	0.10	17.50	70.30	265.92	352240.30
Deferred_tax_liability	2887.00	234.50	2106.25	0.10	3.20	13.50	51.30	72796.60
Shareholders_funds	4256.00	1376.49	13010.69	0.00	32.30	107.60	408.90	613151.60
Cumulative_retained_profits	4211.00	937.18	9853.10	-6534.30	1.10	37.40	206.20	390133.80
Capital_employed	4256.00	2433.62	20496.40	0.00	61.30	221.20	790.30	891408.90
TOL_to_TNW	4256.00	4.03	20.88	-350.48	0.60	1.42	2.83	473.00
Total_term_liabilities__to__tangible_net_worth	4256.00	1.85	15.88	-325.60	0.05	0.34	1.00	456.00
Contingent_liabilities__to__Net_worth_perc	4256.00	55.71	369.17	0.00	0.00	5.36	31.01	14704.27
Contingent_liabilities	2854.00	948.55	12056.74	0.10	6.00	37.85	195.32	559506.80
Net_fixed_assets	4124.00	1209.49	12502.40	0.00	26.20	93.85	352.82	636604.60
Investments	2541.00	721.87	6793.86	0.00	1.00	8.20	63.80	199978.60
Current_assets	4176.00	1350.36	10155.57	0.10	36.60	148.35	515.00	354815.20
Net_working_capital	4219.00	162.87	3182.03	-63839.00	-1.10	16.70	86.50	85782.80
Quick_ratio_times	4151.00	1.50	9.33	0.00	0.41	0.67	1.03	341.00
Current_ratio_times	4151.00	2.26	12.48	0.00	0.93	1.23	1.72	505.00
Debt_to_equity_ratio_times	4256.00	2.87	15.60	0.00	0.22	0.79	1.75	456.00
Cash_to_current_liabilities_times	4151.00	0.53	4.80	0.00	0.02	0.07	0.19	165.00
Cash_to_average_cost_of_sales_per_day	4156.00	145.16	2521.99	0.00	2.88	8.04	21.97	128040.76
Creditors_turnover	3865.00	16.81	75.67	0.00	3.72	6.17	11.69	2401.00
Debtors_turnover	3871.00	17.93	90.16	0.00	3.81	6.47	11.85	3135.20
Finished_goods_turnover	3382.00	84.37	562.64	-0.09	8.19	17.32	40.01	17947.60
WIP_turnover	3492.00	28.68	169.65	-0.18	5.10	9.86	20.24	5651.40
Raw_material_turnover	3828.00	17.73	343.13	-2.00	3.02	6.41	11.82	21092.00
Shares_outstanding	3446.00	23764909.56	170979041.33	-2147483647.00	1308382.50	4750000.00	10906020.00	4130400545.00
Equity_face_value	3446.00	-1094.83	34101.36	-999998.90	10.00	10.00	10.00	100000.00
EPS	4256.00	-196.22	13061.95	-843181.82	0.00	1.49	10.00	34522.53
Adjusted_EPS	4256.00	-197.53	13061.93	-843181.82	0.00	1.24	7.62	34522.53
Total_liabilities	4256.00	3573.62	30074.44	0.10	91.30	315.50	1120.80	1176509.20
PE_on_BSE	1629.00	55.46	1304.45	-1116.64	2.97	8.69	17.00	51002.74

Table 3. Statistical Summary

The following observations were made:

- **Net Worth next year**: The net worth for the next year shows significant variation, with values ranging from negative (indicating a financial loss) to positive. The average net worth is 1,345, with the 75th percentile at 330, and the highest recorded value is 805,773. This suggests a large variation in net worth, with most companies reporting values below 330, but some reaching as high as 805,773.

- **Net Worth**: Notably, the present year's net worth does not show any negative values, indicating that no companies are operating at a loss this year.

- **Shareholders' Funds**: The funds from shareholders range from a minimum of 0 to a 75th percentile value of 408 equities, with a maximum of 613,151. Additionally, there are companies with no reported shareholders' funds in the dataset.

- **Profit After Tax**: Profit after tax shows a range from -3,908 to 119,439, indicating that some companies are experiencing financial losses while others are highly profitable.

- **Cash Profit**: Cash profit, which represents liquidity, ranges from -2,245 to 176,911. The average cash profit is 408, suggesting that while most companies have limited liquidity, there are a few with significant cash flow.

- **Sales and Income**: On average, the companies report sales of 4,645, with a maximum value of 2,384,984. The total income has an average of 4,688, reaching up to 2,442,828, and the total expenses range from 4,356 to 2,366,035. This implies a wide variability in the financial performance of companies, with some companies reporting very high sales and income, while others remain at much lower levels.

- **Liabilities and Borrowings**: The total liabilities average 3,573, with values ranging from 0 to 1,176,509. The 75th percentile for liabilities is under 1,120, indicating that most companies have relatively low liabilities. Borrowings range from 0 to 278,257.

- **Debt-to-Equity Ratio**: The debt-to-equity ratio, which measures the company's leverage, ranges from 0 to 456, with an average of 2.87. This suggests that most companies have relatively low leverage, but some have significantly high levels of debt compared to equity.

- **Current Liabilities and Provisions**: On average, current liabilities and provisions are 960, with a maximum value of 352,240, suggesting variability in short-term obligations across companies.

- **Quick Ratio**: The quick ratio, which measures a company's ability to cover its short-term liabilities without relying on inventory, ranges from 0 to 321, with an average of 1.5. This indicates that while most companies can meet their short-term obligations, a few companies have very low liquidity, suggesting potential concerns about their immediate financial stability.

- **Current Ratio**: The current ratio, which assesses a company's ability to pay off its short-term liabilities with its short-term assets, ranges from 0 to 509, with an average of 2.2. This suggests that, on average, companies have more than enough current assets to cover their liabilities.

- **Cash to Current Liabilities Ratio**: The cash to current liabilities ratio, which indicates the proportion of cash available to meet current liabilities, ranges from 0 to 165, with an average of 0.53. This suggests that, on average, companies have less than one-half of their current liabilities covered by cash alone, indicating potential liquidity risks for some businesses, especially in times of financial stress.

4.5. Univariate analysis

4.5.1. Net worth Next Year

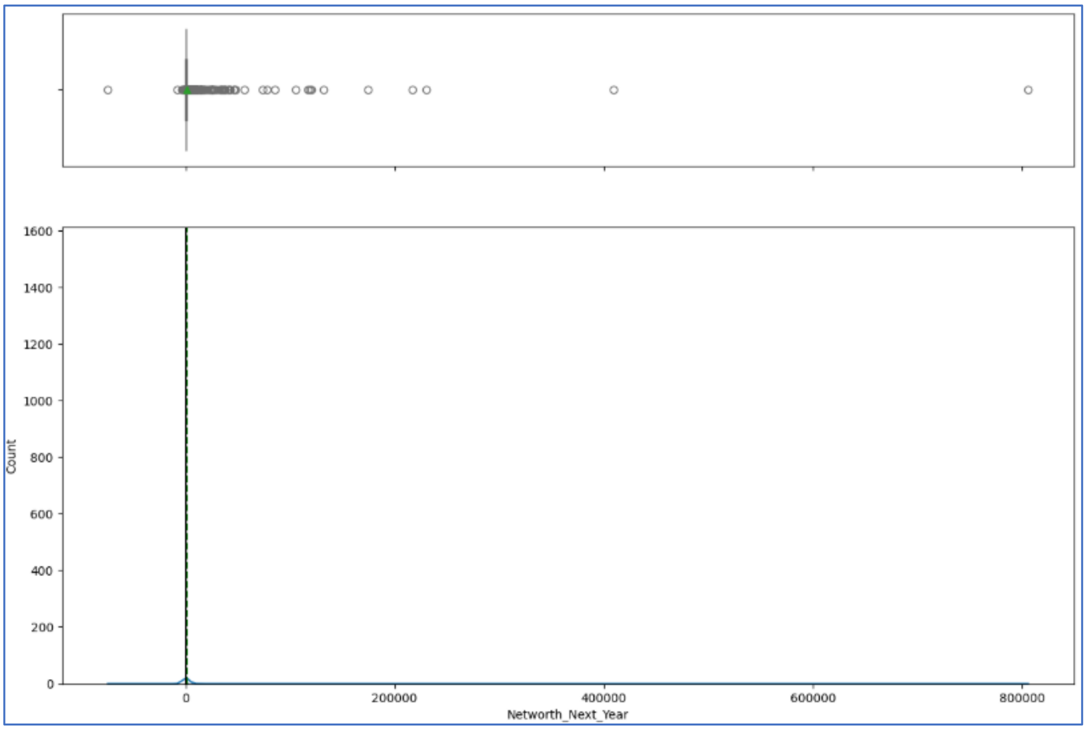

Fig 1. Net worth next year

OBSERVATIONS:

- The projected net worth for the upcoming year exhibits considerable variability, with values spanning from negative (indicating potential financial losses) to positive, with the highest recorded value reaching 800000.
- The data contains a significant number of outliers, which could distort the analysis.
- The mean net worth is 1344, while the standard deviation is 15936, suggesting that most values are dispersed far from the mean, indicating high variability in the data.

4.5.2. Net worth

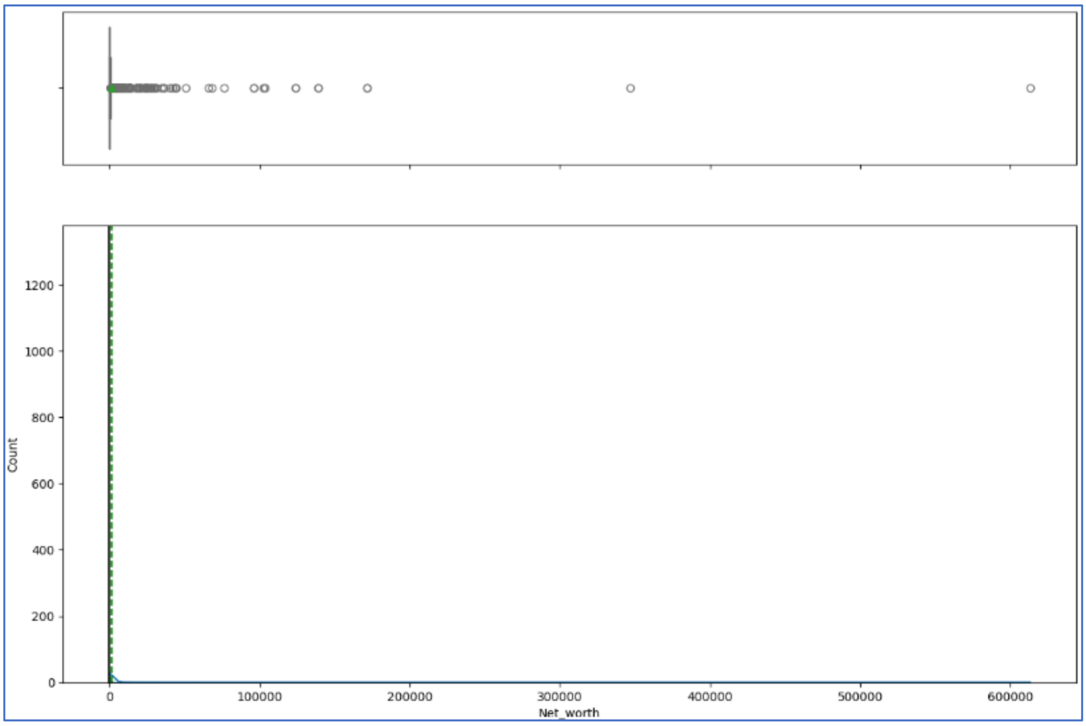

Fig 2. Net worth

OBSERVATIONS:

- The net worth for the current year is entirely positive, with no companies reporting losses.
- The values range from 0 to 600000, but the data contains a significant number of outliers.
- The mean net worth is 1351, with a standard deviation of 12961, indicating considerable variation in the data. The 75th percentile is 389, suggesting that most companies have a net worth below this value.

4.5.3. Total assets

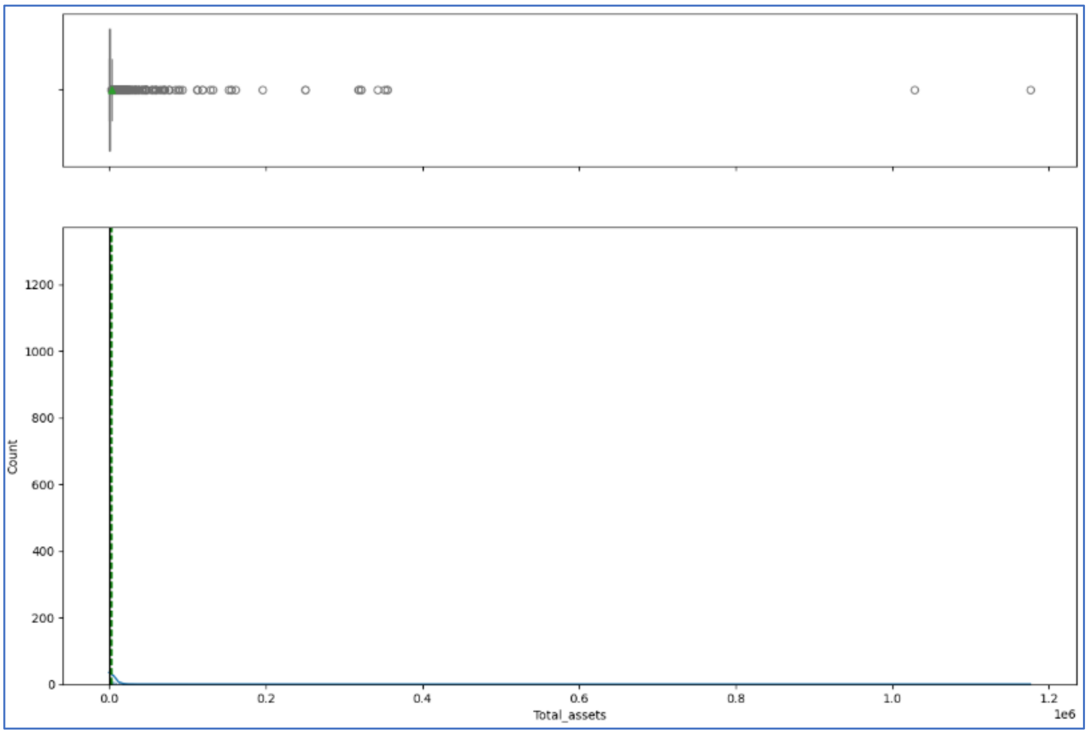

Fig 3. Total assets

OBSERVATIONS:

- The assets range from 0 to 1176509, with several outliers present in the data.
- The distribution is right-skewed, meaning that most values are concentrated on the lower end, with a few extremely high values creating a tail to the right.
- The mean asset value is 3573, and 50% of customers have assets below 315.
- This suggests that most customers hold relatively low asset values, while a small proportion have significantly higher assets, contributing to the right-skewed distribution.

4.5.4. Shareholder’s funds

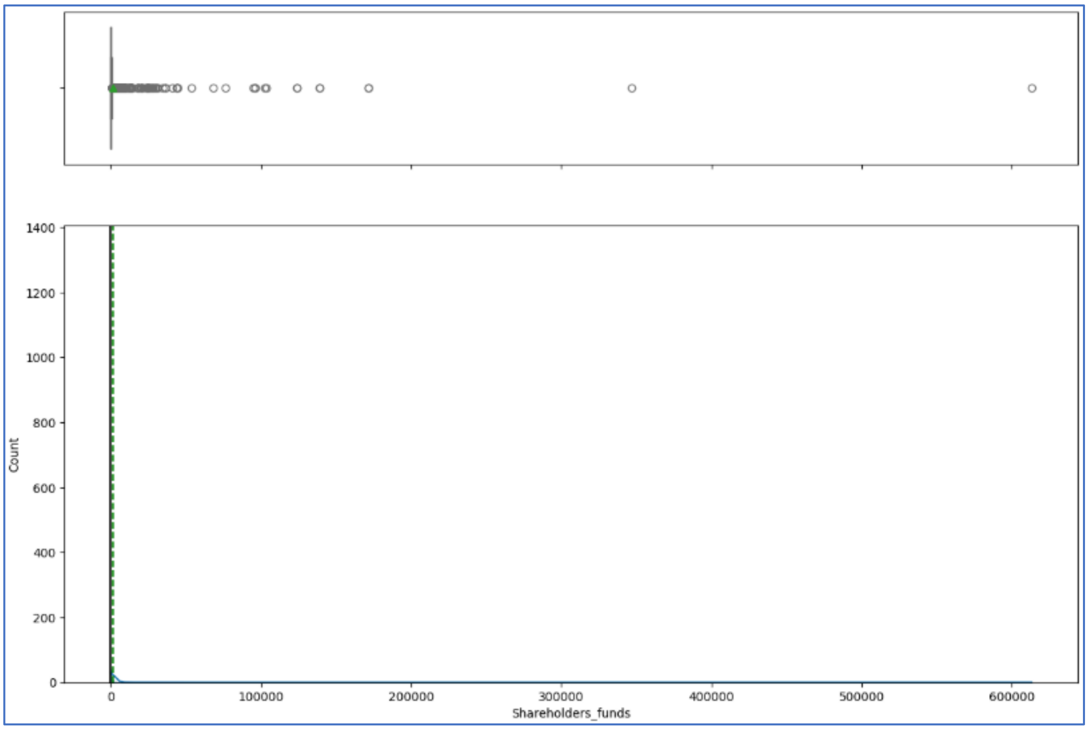

Fig 4. Shareholder’s funds

OBSERVATIONS:

- The shareholders' funds range from 0 to 600000, with a concentration of outliers around the 100,000 marks.
- The distribution is right-skewed, indicating that most values are clustered toward the lower end, while a few values significantly higher create a tail to the right.
- This suggests that most companies have relatively low shareholder funds, with a small number of companies having substantially higher values, which is likely contributing to the right-skewed nature of the distribution.

4.5.5. Investments

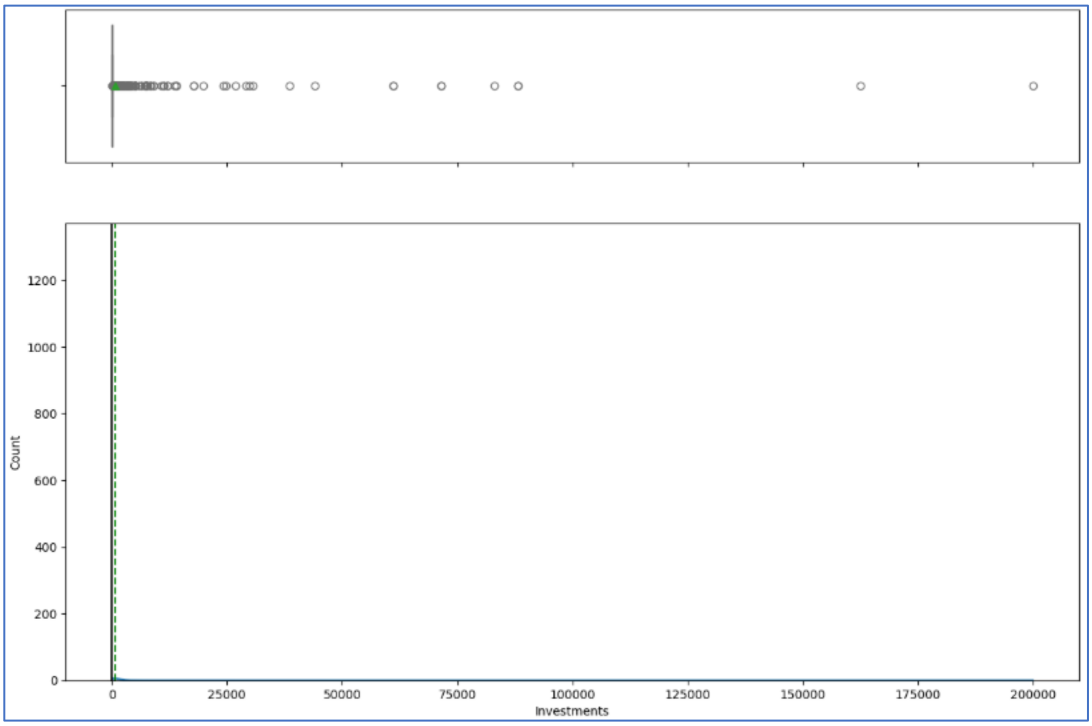

Fig 5. Investments

OBSERVATIONS:

- The investment values exhibit substantial variability, ranging from 0 to 200000, with a significant number of outliers present.
- The distribution is right-skewed, indicating that most values are concentrated at the lower end, with a few extremely high values extending the tail to the right.
- The mean investment is 721, and the highest recorded investment is 190000.
- Additionally, there are a considerable number of missing values in the dataset, which may impact the analysis and require attention.

4.5.6. Total liabilities

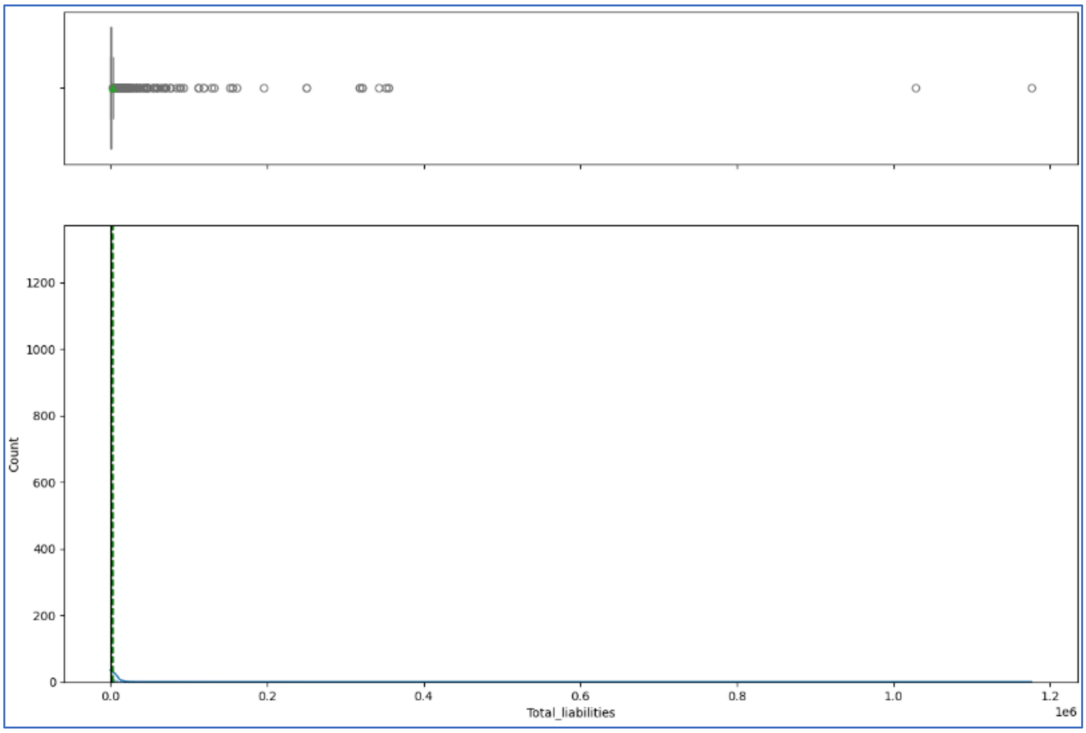

Fig 6. Total liabilities

OBSERVATIONS:

- The total liabilities range from 0 to 1176509, with a significant number of outliers on the higher end. The liabilities span a large range, indicating a broad variation in the financial obligations across the entities in the dataset.
- The presence of outliers on the higher end suggests that a few entities have disproportionately large liabilities compared to the majority.
- Despite the average liability being relatively low at 3573, the large standard deviation of 30,074 indicates that most entities have liabilities significantly different from the average.

4.5.7. Borrowings

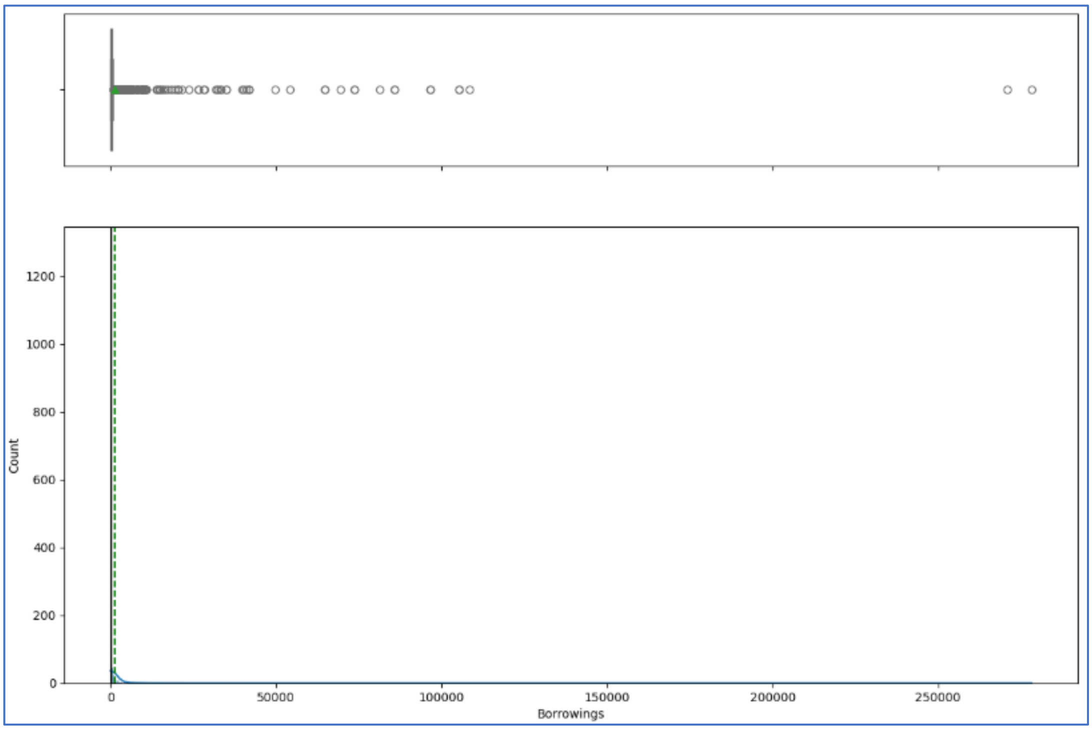

Fig 7. Borrowings

OBSERVATIONS:

- The borrowings range from 0 to 250000, with a significant number of outliers on the higher end of the distribution.
- The dataset contains a considerable number of missing values.
- The 75th percentile of 358 is much lower than the average borrowing amount of 1,176, implying that while most entities have relatively small borrowings, the average is skewed by a few entities with very high borrowing amounts.

4.5.8. Total income

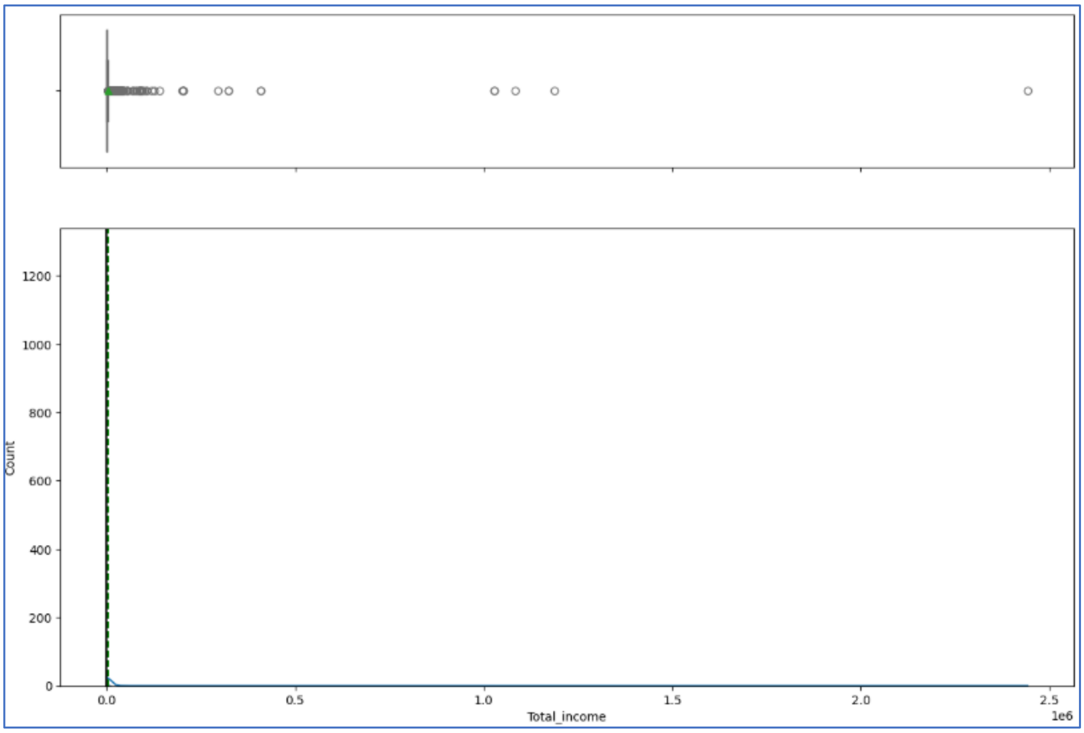

Fig 8. Total income

OBSERVATIONS:

- The total income ranges from 0 to 2442828, with several outliers on the higher end of the distribution. The large range indicates significant variability in income levels across the entities in the dataset.
- The right-side outliers suggest that a few entities are generating exceptionally high incomes compared to the majority, which could be attributed to large-scale operations or exceptional performance by certain entities.
- The standard deviation of 53918 is considerably high compared to the average income of 4688, indicating that most entities have incomes that differ greatly from the average.

4.5.9. Total expenses

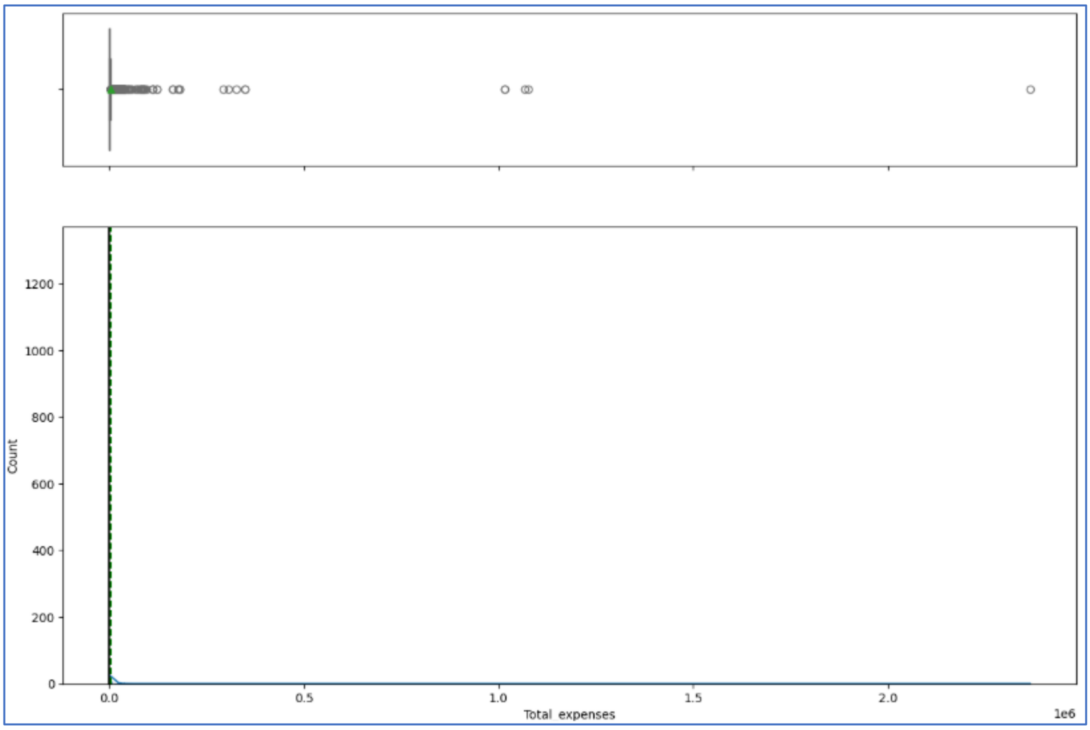

Fig 9. Total expenses

OBSERVATIONS:

- The total expenses vary significantly, with a maximum value of 2366035, indicating a large disparity in the expense levels across the entities in the dataset.
- The median expense of 426 is much lower than the mean of 4,356, suggesting that while most entities have relatively low expenses, the average is influenced by a few entities with significantly higher expenses.
- The higher mean compared to the median indicates a positive skew in the data, meaning that the distribution has a long tail of higher expense values due to a few entities with much larger expenses.

4.5.10. Profit after tax

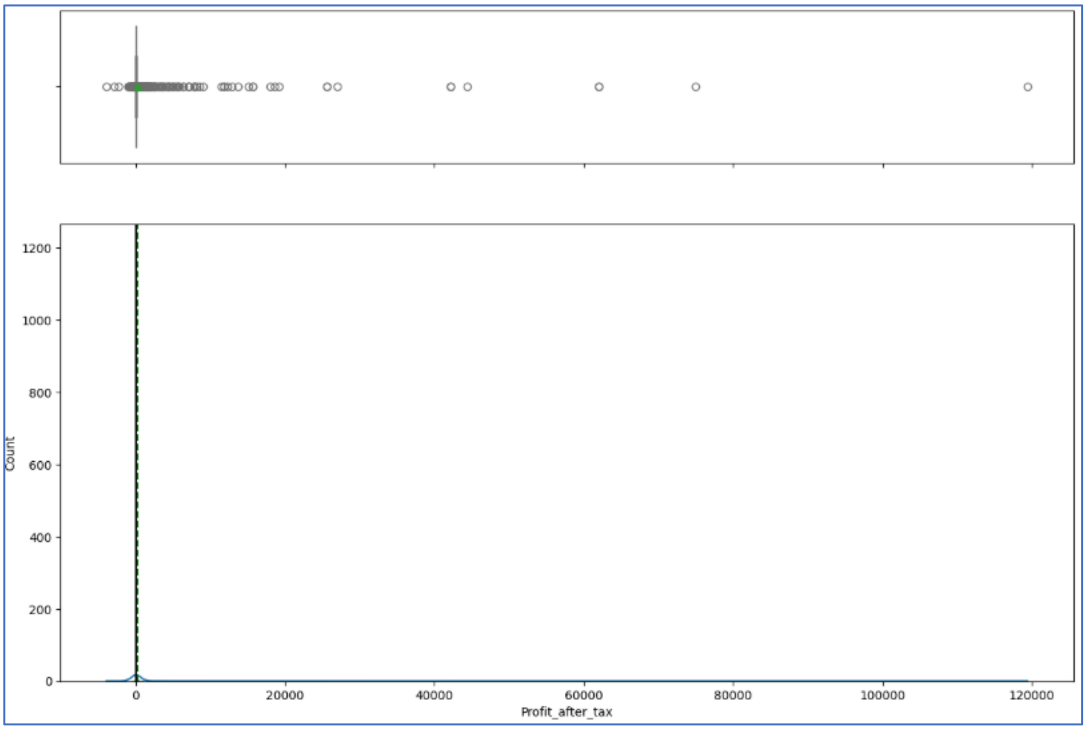

Fig 10. Profit after tax

OBSERVATIONS:

- The profit after tax varies significantly, with values ranging from a negative figure (-3,908) to a high of 119,439, indicating considerable variation in profitability across the entities.
- The negative value suggests that some entities are experiencing losses, which is important for identifying financial distress or struggling companies.
- The 75th percentile of 53 is much lower than the mean of 295, implying that most entities are generating relatively small profits, while a few entities with very high profits are skewing the average.
- The presence of high profit values like 119,439 indicates that some companies are performing exceptionally well, which could be masking the struggles of others that are operating at a loss or with much lower profits.

4.5.11. PBDITA

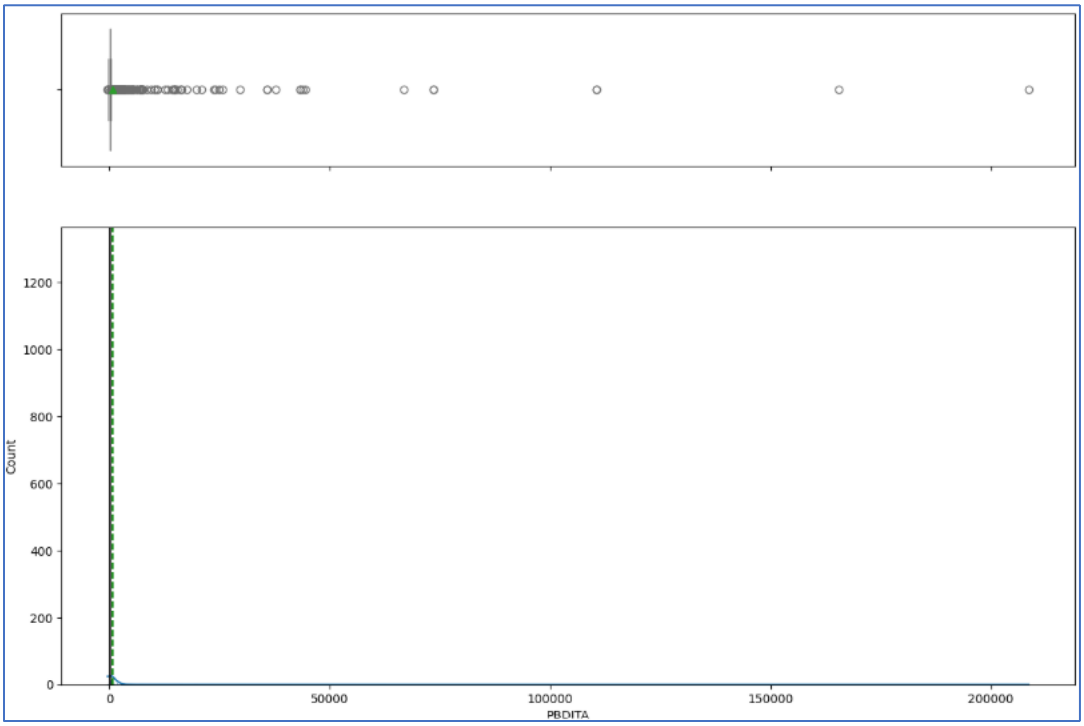

Fig 11. PBDITA

OBSERVATIONS:

- The PBDITA values vary significantly, with a minimum of -440 and a maximum of 208576, suggesting substantial variation in the operational profitability across the entities.
- The negative value of -440 indicates that some entities are facing operational losses, which could be a sign of inefficiencies or financial difficulties in certain companies.
- The median PBDITA of 37 is much lower than the mean of 605, suggesting that while most entities are generating relatively small PBDITA figures, the average is being influenced by a few entities with exceptionally high values.
- The high value of 208576 indicates the presence of a few companies with very high operational profits, which could be distorting the overall picture of the dataset.

4.5.12. Sales

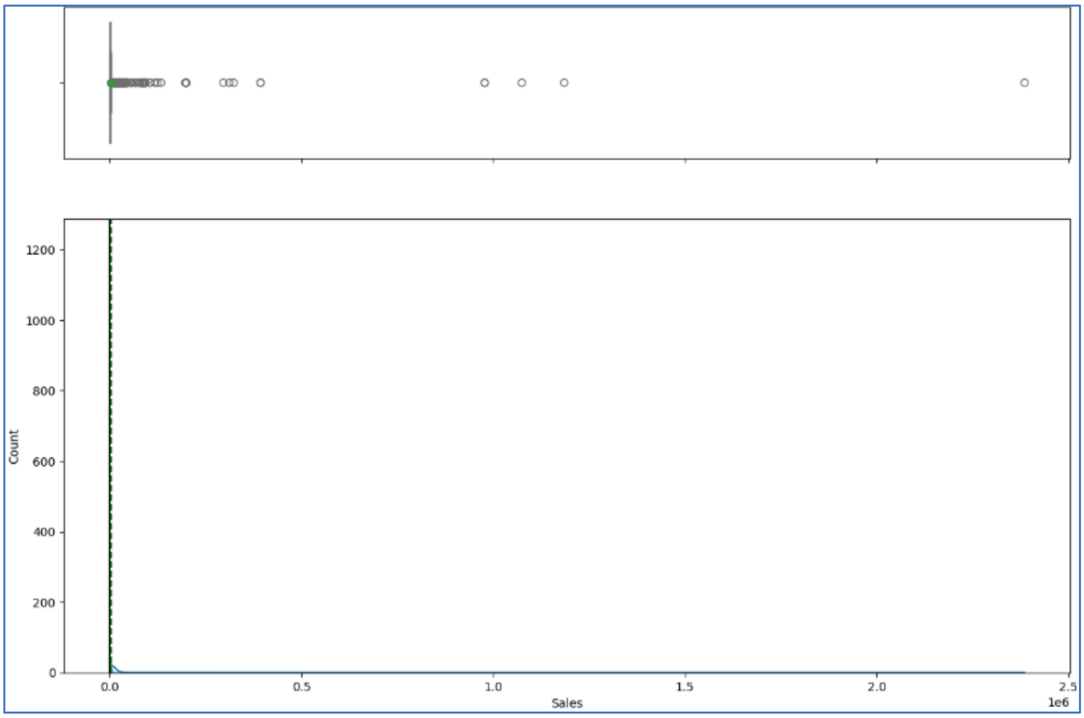

Fig 12. Sales

OBSERVATIONS:

- Sales vary significantly, from 0 to as high as 2384984, indicating that there is a considerable difference in the scale of sales across the entities.
- The lower bound of 0 suggests that some entities have no sales at all, which could point to inactive or struggling companies.
- The standard deviation of 53,080 is quite large compared to the mean of 4,645, indicating that most companies have sales figures that differ widely from the average. This suggests a highly diverse group in terms of sales performance.

4.5.13. Debt to equity ratio times

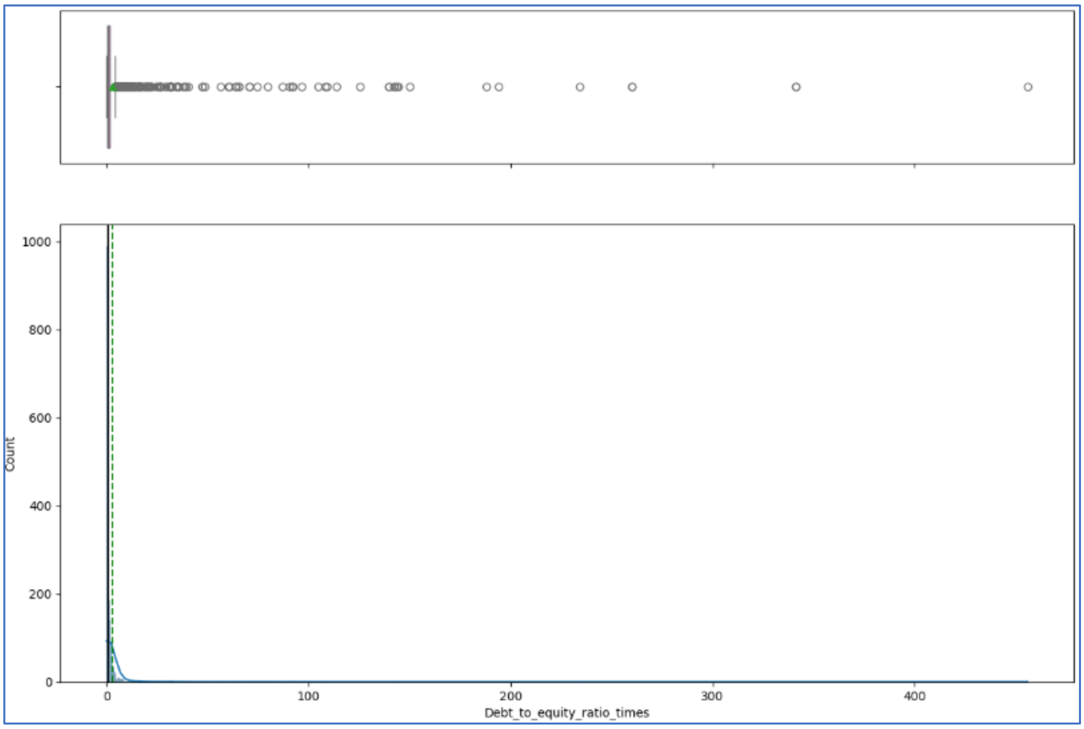

Fig 13. Debt to equity ratio times

OBSERVATIONS:

- The debt-to-equity ratio spans a broad range from 0 to 456, indicating significant variation in the financial leverage used by the entities.
- With the 75th percentile at 1.75, it suggests that most companies have relatively low debt levels compared to equity, indicating conservative financial leverage.
- The average ratio of 2.87 is higher than the 75th percentile, suggesting that while most companies have low to moderate debt levels, a few entities have much higher debt-to-equity ratios that are significantly influencing the average.

4.5.14. Current liabilities & provisions

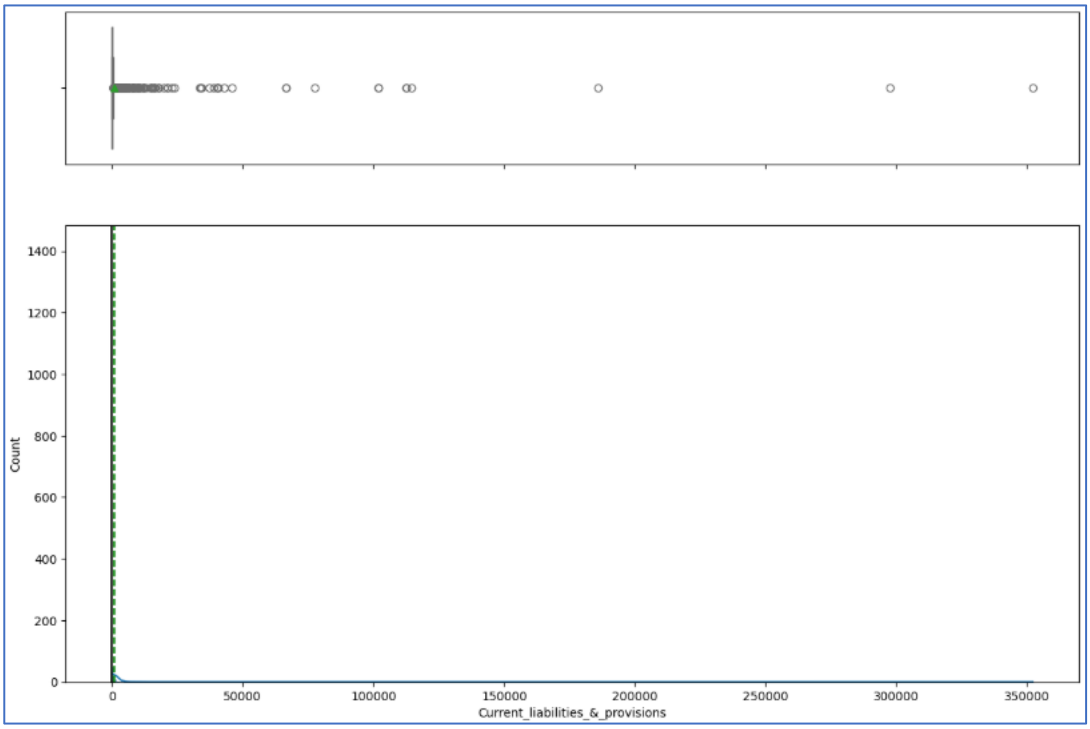

Fig 14. Current liabilities & provisions

OBSERVATIONS:

- The current liabilities and provisions vary greatly, with a maximum value of 352240, indicating significant disparities across the entities in terms of their short-term financial obligations.
- The average value of 960 is relatively low compared to the high standard deviation of 9140, suggesting that most companies have liabilities and provisions that are much lower than the average, but a few companies with very high liabilities are influencing the overall statistics.
- The high standard deviation indicates that a few entities have disproportionately large current liabilities and provisions, which may pose a greater financial burden or risk to those companies.

4.5.15. Quick ratio times

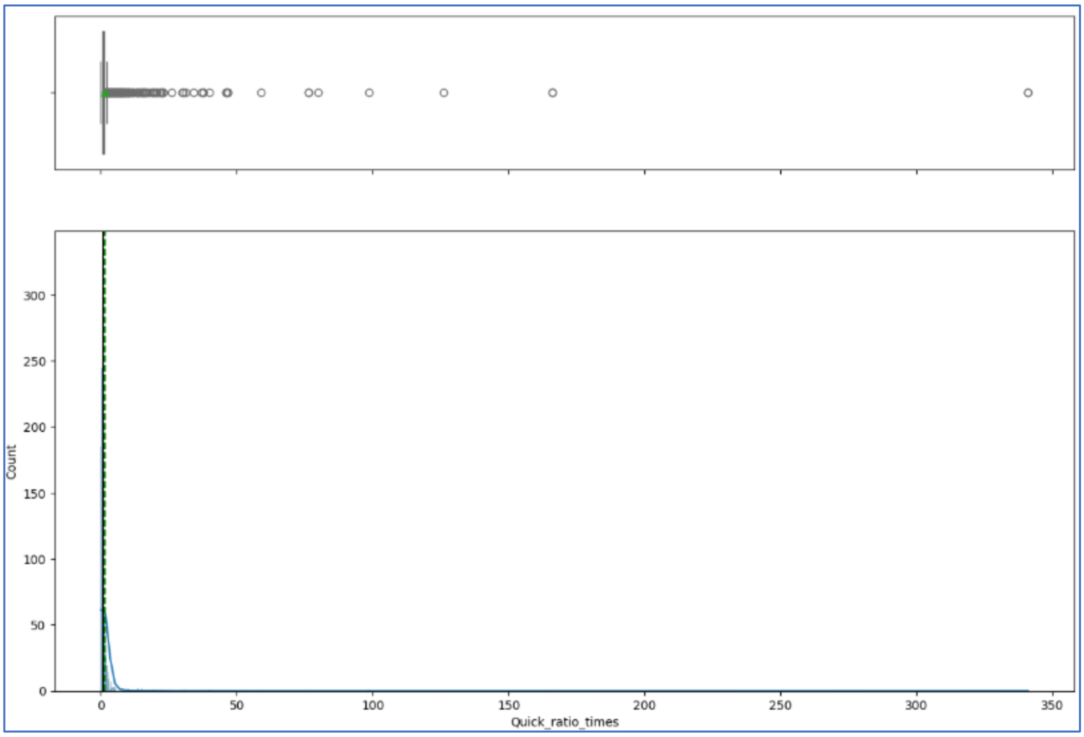

Fig 15. Quick ratio times

OBSERVATIONS:

- The quick ratio varies significantly from 0 to 341, indicating considerable differences in liquidity across the entities in the dataset.
- The median (50th percentile) of 0.67 suggests that half of the entities have a quick ratio below 1, indicating that many companies may not have sufficient liquid assets to cover their short-term liabilities.
- The mean quick ratio of 1.5 is higher than the median, which suggests that a few companies with very high quick ratios are skewing the average. These companies may have an excess of liquid assets relative to their short-term obligations.

4.5.16. Current ratio times

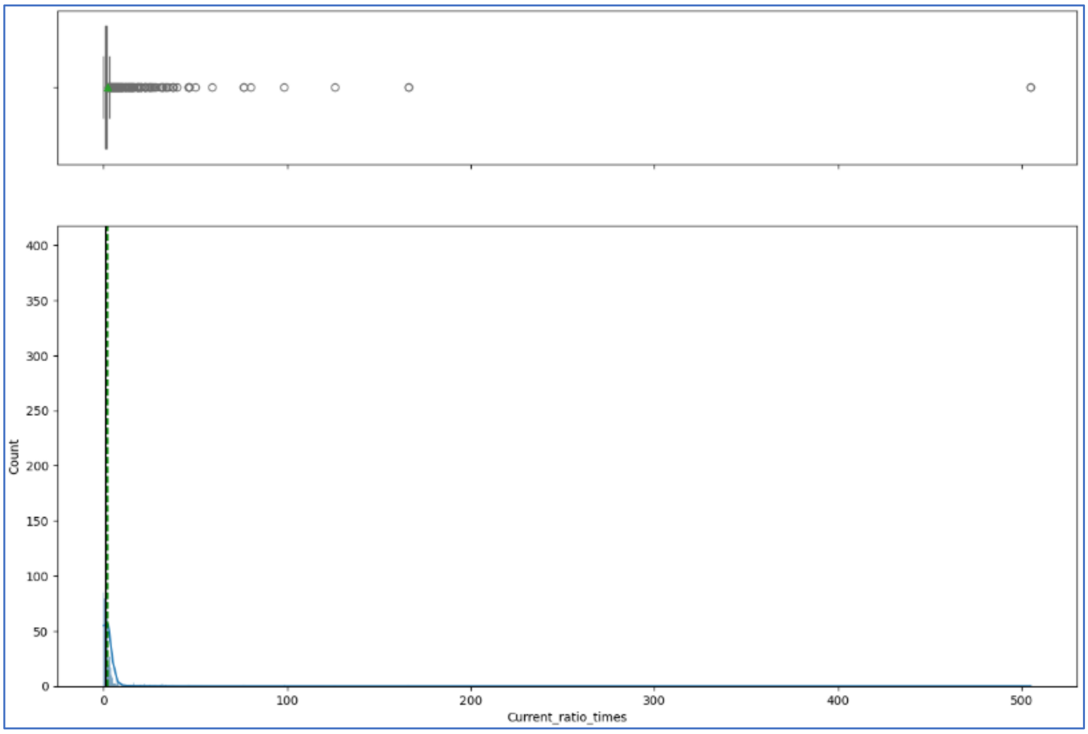

Fig 16. Current ratio times

OBSERVATIONS:

- The current ratio spans from 0 to 505, indicating significant variation in the ability of entities to cover their short-term liabilities with their short-term assets.
- The 75th percentile of 1.72 indicates that the top 25% of companies have even stronger liquidity positions, with more than enough assets to cover their liabilities.
- The maximum current ratio of 505 suggests that some entities may have an excessively high amount of short-term assets relative to their liabilities, potentially indicating over-liquidity or inefficient use of assets.

4.5.17. Cash to current liabilities times

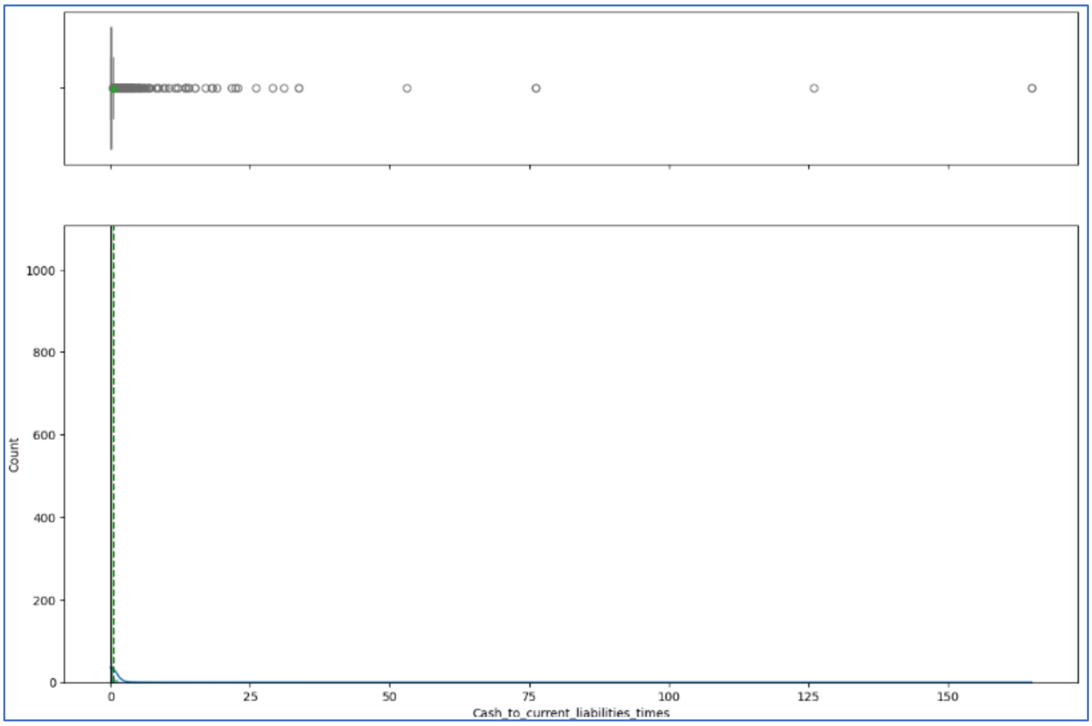

Fig 17. Cash to current liabilities times

OBSERVATIONS:

- The ratio varies significantly from 0 to 165, indicating a large disparity in how companies manage their cash relative to their short-term obligations.
- The median value of 0.07 suggests that most companies have a relatively low amount of cash available to cover their current liabilities, which could indicate potential liquidity concerns for most entities.
- The mean ratio of 0.53 is much higher than the median, implying that a small number of companies have much higher cash reserves relative to their current liabilities, which is skewing the average.
- The maximum value of 165 suggests that a few companies may be holding an unusually large amount of cash relative to their liabilities, which could either indicate strong liquidity management or an inefficient use of cash.

4.5 Bivariate analysis

4.5.1 Net worth Next year vs Total assets

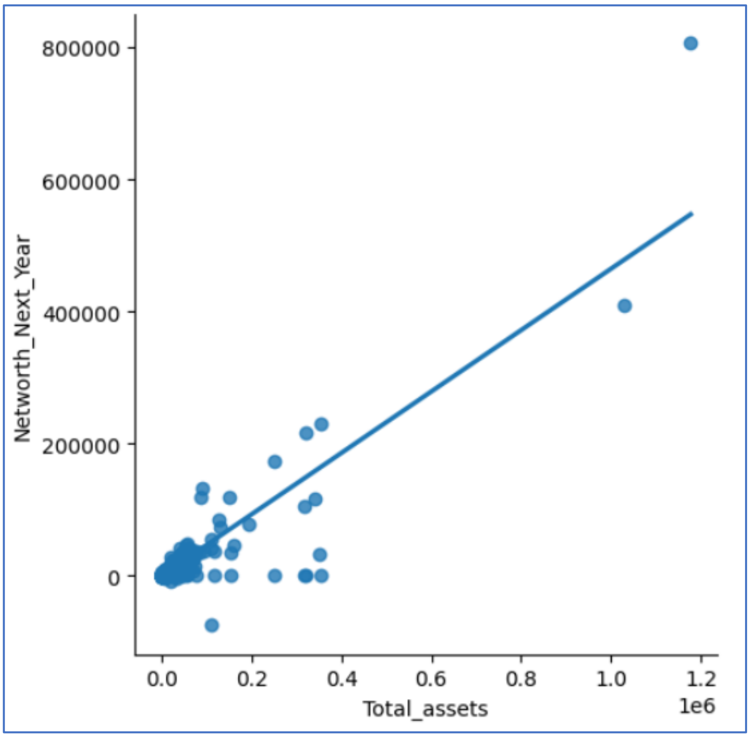

Fig 18. Net worth Next year vs Total assets

OBSERVATIONS:

- The net worth for the next year increases as total assets grow.
- This suggests a positive relationship between total assets and net worth. As companies accumulate more assets, their net worth tends to increase, which could indicate healthy growth and the ability to generate value.

4.5.2 Net worth Next year vs Net worth

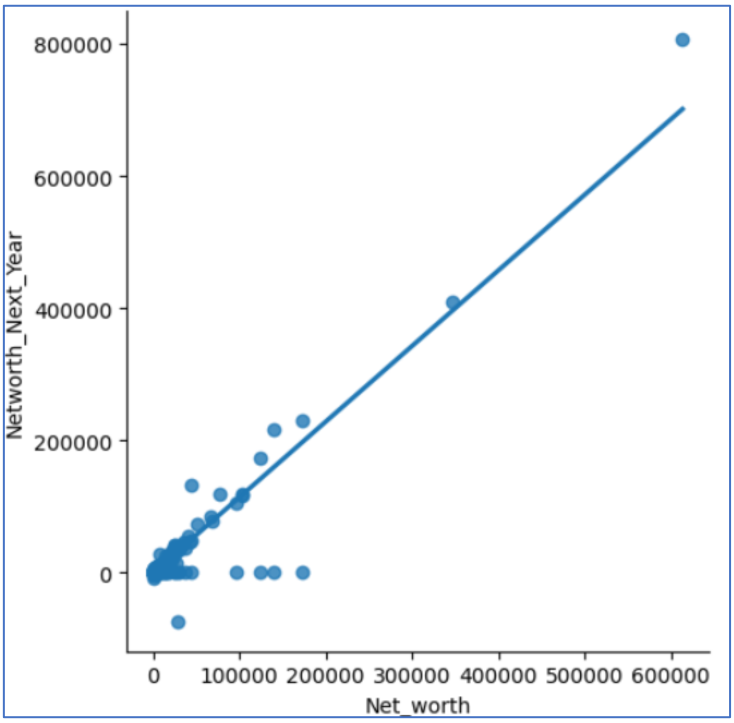

Fig 19. Net worth Next year vs Net worth

OBSERVATIONS:

- The net worth for the next year grows as the net worth of the current year increases.
- This indicates a positive correlation between current and future net worth. When a company's net worth increases in the current year, it is likely to continue growing in the following year.

4.5.3 Net worth Next year vs Total income

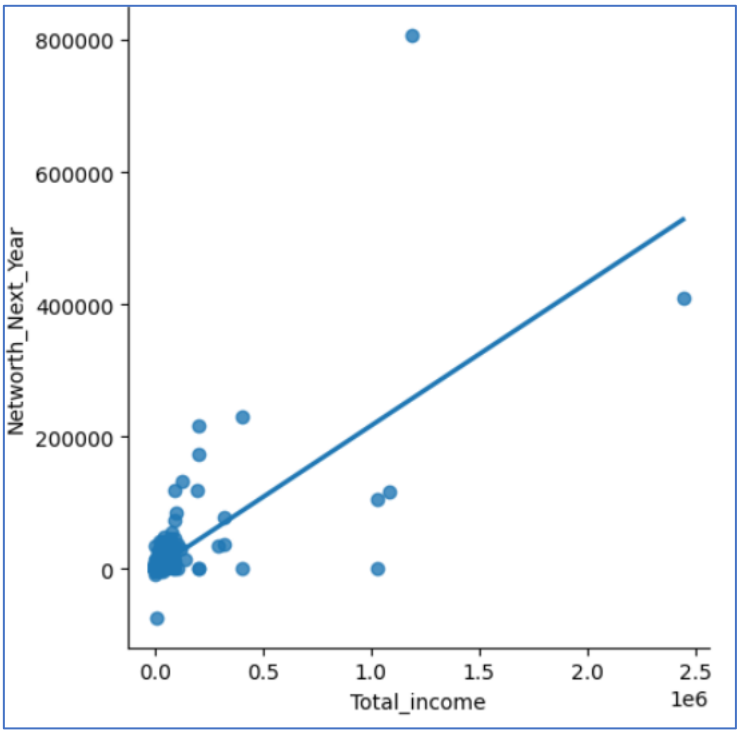

Fig 20. Net worth Next year vs Total income

OBSERVATIONS:

- The net worth for the next year increases as total income rises.
- This indicates that higher total income in the current year tends to lead to greater net worth in the following year, suggesting that profitability plays a key role in building equity.

4.5.4 Net worth Next year vs Profit after tax

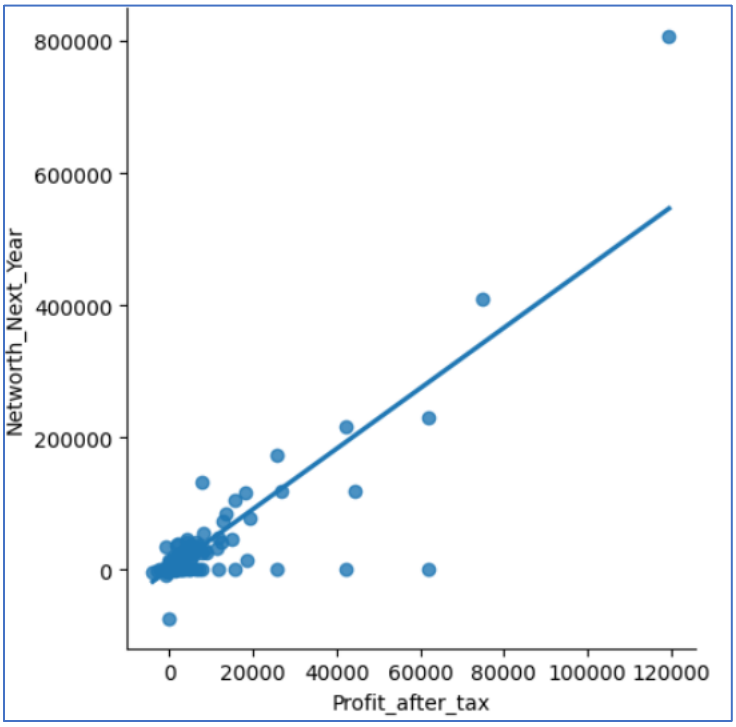

Fig 21. Net worth Next year vs Profit after tax

OBSERVATIONS:

- The net worth for the next year grows as profit after tax increases.
- This suggests that higher profits after tax in the current year contribute directly to the increase in net worth in the following year. Profitable companies can retain earnings, which strengthens their equity base.

4.5.5 Net worth Next year vs Debt to equity ratio times

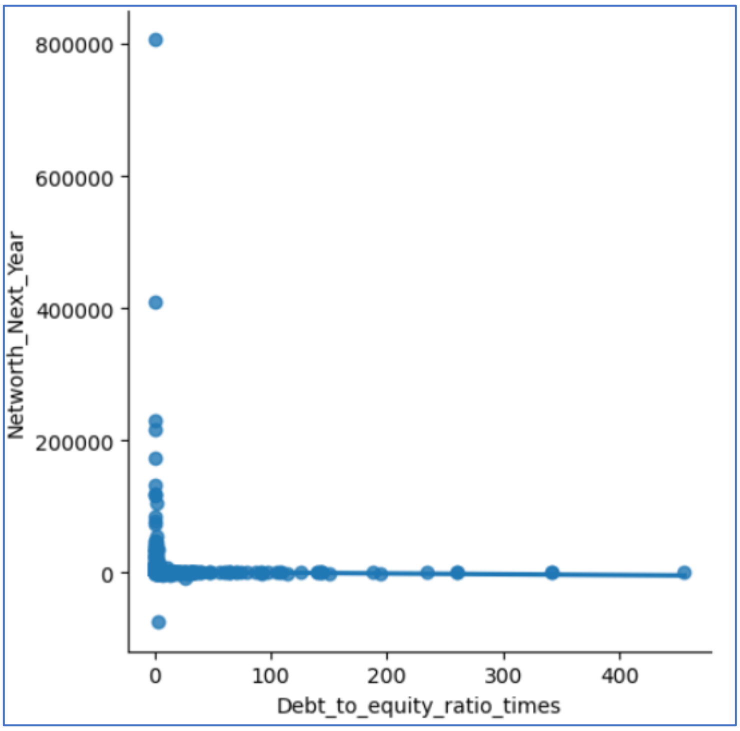

Fig 21. Net worth Next year vs Debt to equity ratio times

OBSERVATIONS:

- The straight-line relationship suggests that variations in the debt-to-equity ratio do not directly influence the net worth for the next year.
- This implies that changes in financial leverage do not necessarily affect the company's overall value or equity position in the short term.

4.5.6 Debt to equity ratio times vs Current liabilities & provisions

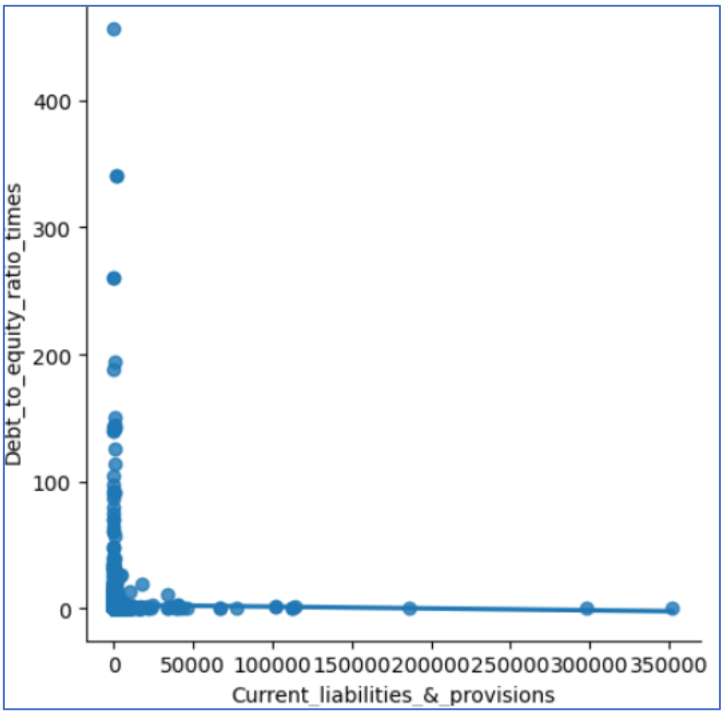

Fig 22. Debt to equity ratio times vs Current liabilities & provisions

OBSERVATIONS:

- The straight-line relationship suggests that variations in the debt-to-equity ratio have no effect on the current liabilities and provisions.
- This implies that the level of debt relative to equity does not influence a company's short-term financial obligations.

4.5.7 Net worth Next year vs Cumulative retained profit

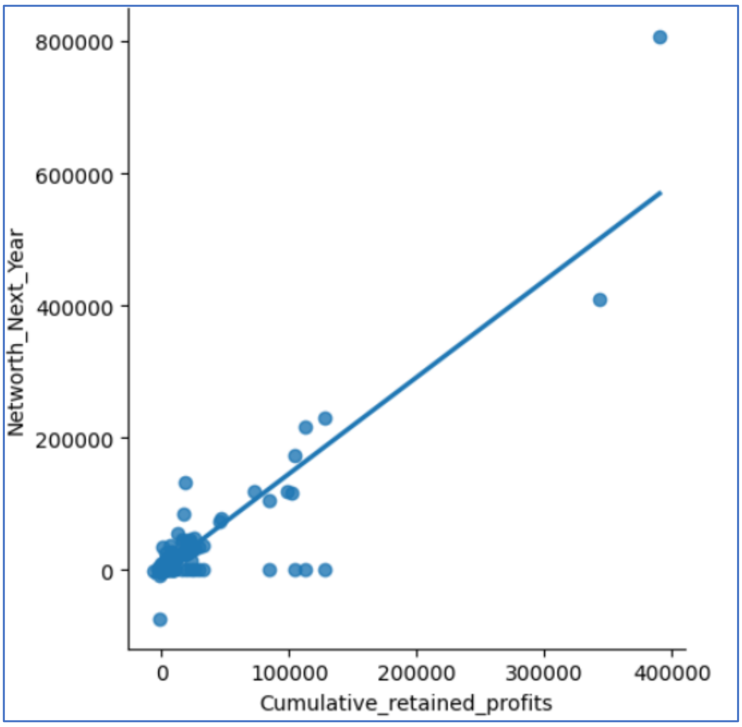

Fig 23. Net worth Next year vs Cumulative retained profit

OBSERVATIONS:

- The net worth for the next year increases as cumulative retained profit grows.
- This indicates that as cumulative retained profits rise, the company's net worth is likely to grow. Retained profits contribute directly to equity, strengthening the company's overall financial position.

4.6 Multivariate analysis

4.6.1 Heat Map

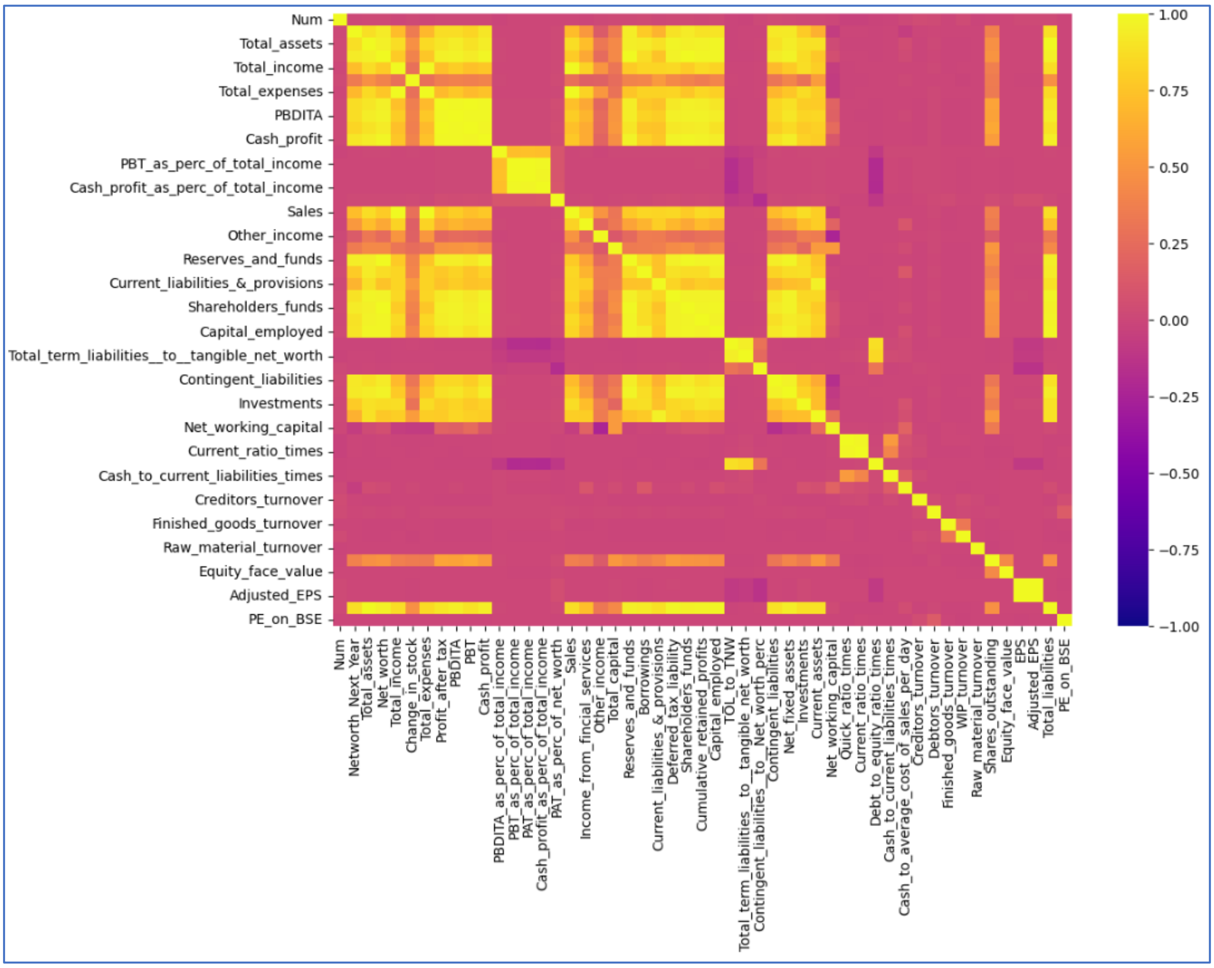

Fig 24. Heat Map

OBSERVATIONS:

- **Positive Correlation**:
  The net worth for the next year shows a strong positive correlation with total assets, total income, total expenses, PBDITA, cash profit, sales, reserves and funds, shareholders' funds, capital employed, and investments. This indicates that increases in these factors generally contribute to higher net worth, suggesting they play a crucial role in driving future financial growth.

- **Negative Correlation**:
  There is a strong negative correlation between net worth next year and factors like cash profit as a percentage of total income, PBT as a percentage of total income, total term liabilities, current ratio, cash to current liabilities, creditors turnover, raw material turnover, and adjusted EPS. This suggests that higher values in these metrics are associated with lower net worth growth, possibly pointing to inefficiencies or financial strains that hinder the company's ability to build equity.

4.7 Insights based on Exploratory Data Analysis

4.7.1 Based on Univariate analysis

- There's a wide range in projected net worth, with some entities showing potential financial struggles while others expect substantial growth, reflecting varied financial health across companies.
- Both asset and shareholders' fund distributions are right-skewed, indicating that most companies have modest holdings, while a few dominate the totals, contributing to significant disparities in financial strength.
- Investments show high variability, with a few companies holding significantly higher amounts than the majority, further skewing the overall distribution.
- There's a large variation in liabilities, with a few companies holding disproportionately high obligations compared to others, suggesting financial stress for some entities.
- Most companies have low borrowings, but a few with high borrowings distort the overall average, indicating differing financial strategies or risks.
- Both income and expenses show considerable variability, with a few companies performing exceptionally well or incurring large expenses, while most remain in a lower range, highlighting diverse business scales.
- Profit after tax and operational profits (PBDITA) also vary greatly, with some companies thriving and others facing financial difficulties, suggesting mixed financial health.
- Sales performance varies widely, with some companies reporting high sales and others with minimal or no sales, reflecting differences in business size and activity.
- The debt-to-equity ratio shows significant variability, with most companies maintaining low debt levels, but a few with very high ratios, pointing to varying degrees of financial risk.
- Current liabilities and provisions exhibit disparities, with most companies having low amounts, but a few having large liabilities, potentially affecting financial stability.
- Quick and current ratios highlight liquidity differences across entities, with many companies potentially facing liquidity challenges, while others maintain strong liquidity or may be inefficient in cash utilization.
- The cash to current liabilities ratio shows disparities, with most companies holding limited cash reserves compared to liabilities, indicating potential liquidity risks for the majority, while a few maintain stronger liquidity positions.

4.7.2 Based on Bivariate analysis

- As total assets increase, so does the net worth, suggesting that companies with more assets tend to build higher equity.
- A positive correlation indicates that companies experiencing growth in the current year's net worth are likely to see continued growth in the next year.
- Higher income in the current year generally leads to a stronger net worth, highlighting the importance of profitability in increasing equity.
- Companies with higher profits are more likely to see a rise in net worth, as retained earnings bolster their equity.
- As cumulative retained profits increase, so does net worth, showing that retaining earnings contributes directly to financial growth.
- A straight-line relationship suggests that variations in financial leverage do not significantly impact net worth in the short term.
- Similarly, changes in the debt-to-equity ratio do not appear to influence a company's short-term financial obligations.
- A strong negative correlation exists between net worth and factors such as cash profit as a percentage of total income, PBT as a percentage of total income, total term liabilities, and liquidity ratios. This suggests that higher values in these metrics may indicate inefficiencies or financial strains that hinder net worth growth.

5. Data Preprocessing

5.1. Target variable creation using 'Networth_Next_Year'

The target variable, 'Networth_Next_Year', is created as a binary variable, where a value of 0 represents positive net worth for the next year, and a value of 1 represents negative net worth for the next year.

	default	Networth_Next_Year
0	0	395.30
1	0	36.20
2	0	84.00
3	0	2041.40
4	0	41.80
5	0	291.50
6	0	93.30
7	0	985.10
8	0	188.60
9	0	229.60
10	0	292.80
11	0	12300.00
12	0	3232.30
13	1	-5.30
14	0	26.70

Table 4. Target variable creation

5.2. Proportion of default

The proportion of the binary variable in the dataset is calculated, revealing that 79% of companies have a positive net worth for the next year (denoted by 0), while 21% have a negative net worth (denoted by 1). This indicates that the dataset is imbalanced.

```
Company['default'].value_counts()

default
0    3352
1     904
Name: count, dtype: int64

Company['default'].value_counts(normalize = True)

default
0   0.79
1   0.21
Name: proportion, dtype: float64
```

Table 5. Proportion of default

5.3. Missing value check

The dataset contains a significant number of missing values, which need to be imputed in order to achieve accurate and reliable results.

```
Num                                               0
Networth_Next_Year                                0
Total_assets                                      0
Net_worth                                         0
Total_income                                    231
Change_in_stock                                 550
Total_expenses                                  165
Profit_after_tax                                154
PBDITA                                          154
PBT                                             154
Cash_profit                                     154
PBDITA_as_perc_of_total_income                   79
PBT_as_perc_of_total_income                      79
PAT_as_perc_of_total_income                      79
Cash_profit_as_perc_of_total_income              79
PAT_as_perc_of_net_worth                          0
Sales                                           305
Income_from_fincial_services                   1111
Other_income                                   1556
Total_capital                                     5
Reserves_and_funds                               98
Borrowings                                      431
Current_liabilities_&_provisions                110
Deferred_tax_liability                         1369
Shareholders_funds                                0
Cumulative_retained_profits                      45
Capital_employed                                  0
TOL_to_TNW                                        0
Total_term_liabilities__to__tangible_net_worth    0
Contingent_liabilities__to__Net_worth_perc        0
Contingent_liabilities                         1402
Net_fixed_assets                                132
Investments                                    1715
Current_assets                                   80
Net_working_capital                              37
Quick_ratio_times                               105
Current_ratio_times                             105
Debt_to_equity_ratio_times                        0
Cash_to_current_liabilities_times               105
Cash_to_average_cost_of_sales_per_day           100
Creditors_turnover                              391
Debtors_turnover                                385
Finished_goods_turnover                         874
WIP_turnover                                    764
Raw_material_turnover                           428
Shares_outstanding                              810
Equity_face_value                               810
EPS                                               0
Adjusted_EPS                                      0
Total_liabilities                                 0
PE_on_BSE                                      2627
default                                           0
dtype: int64
```

Table 6. Missing value

5.4. Outlier check

The dataset contains numerous outliers, which is common in datasets involving companies, especially those under the scrutiny of venture capitalists. These outliers could represent extreme values or atypical behaviour of certain companies, and they need to be addressed appropriately.

Proper handling of outliers, whether through removal, transformation, or other techniques, will ensure that the results are more representative of typical company performance and improve the overall accuracy of the analysis.

```
Num                                            0
Networth_Next_Year                           624
Total_assets                                 585
Net_worth                                    595
Total_income                                 508
Change_in_stock                              750
Total_expenses                               518
Profit_after_tax                             712
PBDITA                                       584
PBT                                          704
Cash_profit                                  627
PBDITA_as_perc_of_total_income               346
PBT_as_perc_of_total_income                  546
PAT_as_perc_of_total_income                  610
Cash_profit_as_perc_of_total_income          426
PAT_as_perc_of_net_worth                     427
Sales                                        500
Income_from_fincial_services                 517
Other_income                                 389
Total_capital                                551
Reserves_and_funds                           643
Borrowings                                   532
Current_liabilities_&_provisions             581
Deferred_tax_liability                       406
Shareholders_funds                           588
Cumulative_retained_profits                  699
Capital_employed                             572
TOL_to_TNW                                   414
Total_term_liabilities__to__tangible_net_worth 406
Contingent_liabilities__to__Net_worth_perc   478
Contingent_liabilities                       393
Net_fixed_assets                             569
Investments                                  451
Current_assets                               532
Net_working_capital                          806
Quick_ratio_times                            371
Current_ratio_times                          397
Debt_to_equity_ratio_times                   381
Cash_to_current_liabilities_times            539
Cash_to_average_cost_of_sales_per_day        583
Creditors_turnover                           442
Debtors_turnover                             408
Finished_goods_turnover                      399
WIP_turnover                                 378
Raw_material_turnover                        296
Shares_outstanding                           476
Equity_face_value                            533
EPS                                          638
Adjusted_EPS                                 694
Total_liabilities                            585
PE_on_BSE                                    237
dtype: int64
```

Table 7. Outliers

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5.5. Converting outliers to Nan

Converting outliers to NaN is an approach used to handle outliers in a dataset. By replacing extreme values or outliers with NaN, we essentially mark those data points as missing or invalid.

Rather than removing the entire row or column containing the outlier, replacing the outliers with NaN allows us to retain the structure of the dataset.

```
Num                                           0
Networth_Next_Year                            0
Total_assets                                  0
Net_worth                                     0
Total_income                                231
Change_in_stock                             550
Total_expenses                              165
Profit_after_tax                            154
PBDITA                                      154
PBT                                         154
Cash_profit                                 154
PBDITA_as_perc_of_total_income               79
PBT_as_perc_of_total_income                  79
PAT_as_perc_of_total_income                  79
Cash_profit_as_perc_of_total_income          79
PAT_as_perc_of_net_worth                      0
Sales                                       305
Income_from_fincial_services               1111
Other_income                               1556
Total_capital                                 5
Reserves_and_funds                           98
Borrowings                                  431
Current_liabilities_&_provisions            110
Deferred_tax_liability                     1369
Shareholders_funds                            0
Cumulative_retained_profits                  45
Capital_employed                              0
TOL_to_TNW                                    0
Total_term_liabilities__to__tangible_net_worth    0
Contingent_liabilities__to__Net_worth_perc    0
Contingent_liabilities                     1402
Net_fixed_assets                            132
Investments                                1715
Current_assets                               80
Net_working_capital                          37
Quick_ratio_times                           105
Current_ratio_times                         105
Debt_to_equity_ratio_times                    0
Cash_to_current_liabilities_times           105
Cash_to_average_cost_of_sales_per_day       100
Creditors_turnover                          391
Debtors_turnover                            385
Finished_goods_turnover                     874
WIP_turnover                                764
Raw_material_turnover                       428
Shares_outstanding                          810
Equity_face_value                           810
EPS                                           0
Adjusted_EPS                                  0
Total_liabilities                             0
PE_on_BSE                                  2627
default                                       0
dtype: int64
```

Table 8. Total missing values inclusive of outliers converted to Nan

5.6. Visualization of Missing values

The heatmap visually illustrates the missing values in the dataset, providing an overview of which variables have missing data.

The colour coding in the heatmap highlights the presence of missing values, making it easier to identify patterns. It is observed that the columns Other_Income, Deferred_Tax_Liability, Contingent_Liabilities, Investments, and PE_on_BSE have a significant number of missing values.

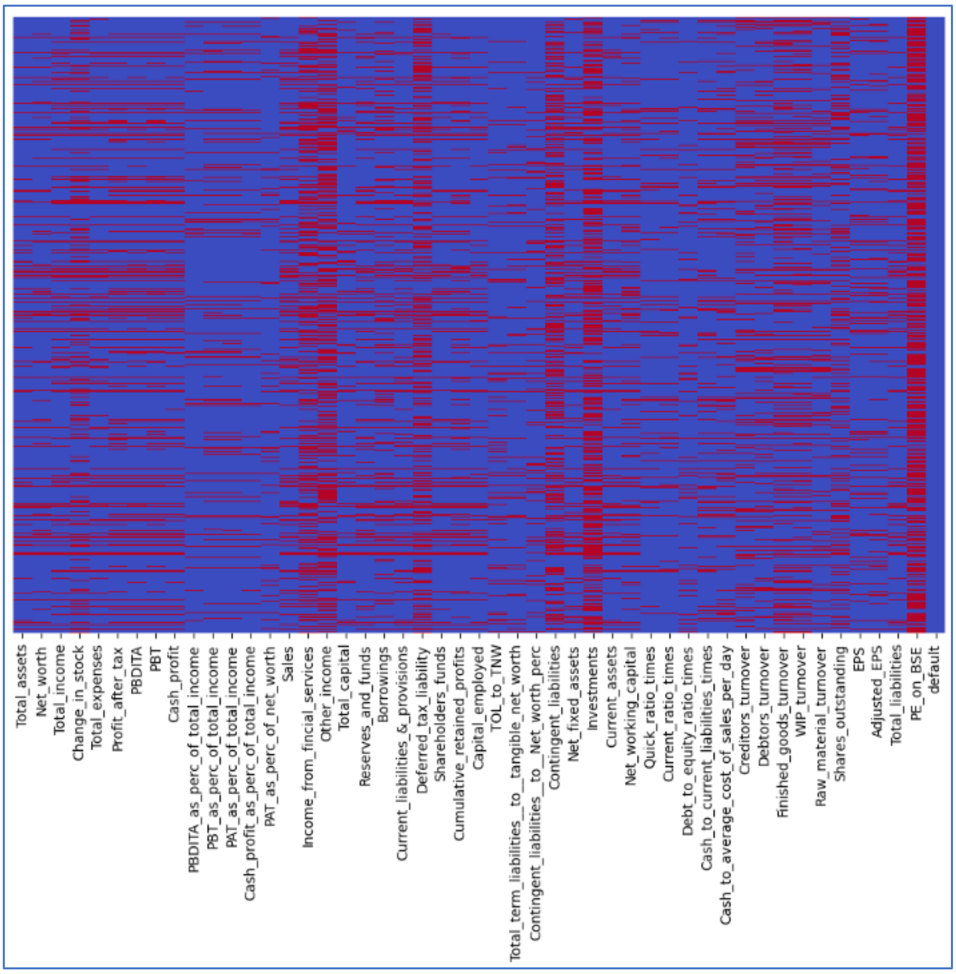

Fig 25. Missing values: Heat Map

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5.7. Percentage of Missing values in each variable

By filtering out rows with less than 90% completeness, we assess the percentage of missing values in the dataset. Since having 30% or more missing data is deemed problematic for analysis, we calculate the proportion of missing values to understand their potential impact on the overall analysis.

```
PE_on_BSE                                     0.67
Investments                                   0.51
Other_income                                  0.46
Contingent_liabilities                        0.42
Deferred_tax_liability                        0.42
Income_from_fincial_services                  0.38
Change_in_stock                               0.31
Shares_outstanding                            0.30
Finished_goods_turnover                       0.30
WIP_turnover                                  0.27
Borrowings                                    0.23
Profit_after_tax                              0.20
PBT                                           0.20
Net_working_capital                           0.20
Creditors_turnover                            0.20
Sales                                         0.19
Debtors_turnover                              0.19
Cash_profit                                   0.18
Cumulative_retained_profits                   0.17
Reserves_and_funds                            0.17
Total_income                                  0.17
PBDITA                                        0.17
Raw_material_turnover                         0.17
Net_fixed_assets                              0.16
Adjusted_EPS                                  0.16
Current_liabilities_&_provisions              0.16
PAT_as_perc_of_total_income                   0.16
Cash_to_average_cost_of_sales_per_day         0.16
Total_expenses                                0.16
Cash_to_current_liabilities_times             0.15
EPS                                           0.15
PBT_as_perc_of_total_income                   0.15
Current_assets                                0.14
Net_worth                                     0.14
Shareholders_funds                            0.14
Total_liabilities                             0.14
Total_assets                                  0.14
Capital_employed                              0.13
Total_capital                                 0.13
Cash_profit_as_perc_of_total_income           0.12
Current_ratio_times                           0.12
Contingent_liabilities__to__Net_worth_perc    0.11
Quick_ratio_times                             0.11
PAT_as_perc_of_net_worth                      0.10
PBDITA_as_perc_of_total_income                0.10
TOL_to_TNW                                    0.10
Total_term_liabilities__to__tangible_net_worth 0.10
Debt_to_equity_ratio_times                    0.09
default                                       0.00
dtype: float64
```

Table 9. Percentage of missing values in each variable

5.8. Missing value imputation

After filtering out the variables that have more than 30% missing values, which can potentially introduce bias and reduce the quality of analysis, we move on to handle the remaining missing data.

To address these remaining gaps, we apply the KNNImputer method, a popular technique for imputing missing values which works by finding the k-nearest neighbours for each data point with missing values and then using the average of those neighbour's values to fill in the missing data.

Scaling the data with StandardScaler is significant because it standardizes the features by transforming them to have a mean of 0 and a standard deviation of 1. This ensures that all features are on the same scale and prevents the model from being biased towards variables with larger numerical ranges.

Total_assets	0
Net_worth	0
Total_income	0
Total_expenses	0
Profit_after_tax	0
PBDITA	0
PBT	0
Cash_profit	0
PBDITA_as_perc_of_total_income	0
PBT_as_perc_of_total_income	0
PAT_as_perc_of_total_income	0
Cash_profit_as_perc_of_total_income	0
PAT_as_perc_of_net_worth	0
Sales	0
Total_capital	0
Reserves_and_funds	0
Borrowings	0
Current_liabilities_&_provisions	0
Shareholders_funds	0
Cumulative_retained_profits	0
Capital_employed	0
TOL_to_TNW	0
Total_term_liabilities__to__tangible_net_worth	0
Contingent_liabilities__to__Net_worth_perc	0
Net_fixed_assets	0
Current_assets	0
Net_working_capital	0
Quick_ratio_times	0
Current_ratio_times	0
Debt_to_equity_ratio_times	0
Cash_to_current_liabilities_times	0
Cash_to_average_cost_of_sales_per_day	0
Creditors_turnover	0
Debtors_turnover	0
Finished_goods_turnover	0
WIP_turnover	0
Raw_material_turnover	0
Shares_outstanding	0
EPS	0
Adjusted_EPS	0
Total_liabilities	0
default	0
dtype: int64	

Table 10. Treated missing values and outliers

5.9. Correlation between independent variables

High correlation between independent variables is problematic in a dataset as it can lead to multicollinearity, which distorts the model's performance.

The heatmap below illustrates the relationships between independent variables, where total assets show a strong positive correlation with net worth, total income, total expenses, profit after tax, and PBDITA.

These correlations need to be addressed to reduce multicollinearity before fitting the model, ensuring more accurate and stable results.

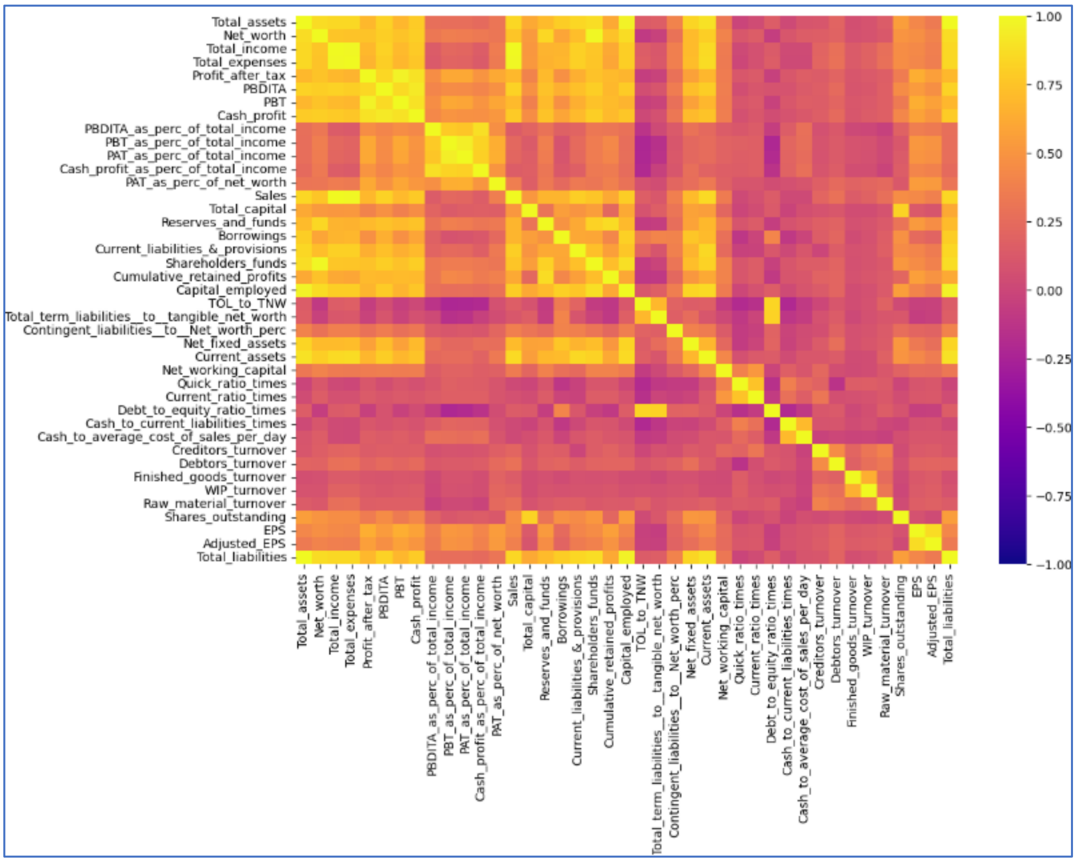

Fig 26. Independent variables HeatMap

5.11. Data preparation

The data set has been split into training and test set with a split of 70% and 30% respectively.

```
Shape of Training set :  (2979, 41)
Shape of test set :  (1277, 41)
Percentage of classes in training set:
default
0.00   0.79
1.00   0.21
Name: proportion, dtype: float64
Percentage of classes in test set:
default
0.00   0.78
1.00   0.22
Name: proportion, dtype: float64
```

Table 11. Train-Test Data Split

Around 79% of observations belongs to class 0 (non-default) and 21% observations belongs to class 1 (default), and this is preserved in the train and test sets.

6. Model Building

The model can make wrong predictions as:

- Predicting negative values in Networth_Next_Year will not happen but in reality, the negative values will happen (False negative, FN)

- Predicting negative values in Networth_Next_Year will happen but in reality, the negative values will not happen (False positive, FP)

The aim is to reduce false negatives, which can be done by maximizing the Recall. Greater the recall lesser the chances of false negatives.

6.1. Metrics of Choice

- **Recall:** It is the ability of the model to correctly identify positive cases (in this case, predicting Networth_Next_Year as negative when it is actually negative). In the case of Networth_Next_Year, false negatives represent missing out on companies that are actually facing a financial downturn but are incorrectly predicted to have a positive net worth. A higher recall means fewer instances where the model misses predicting negative values (FN), thus reducing false negatives.

- **Accuracy:** It measures the proportion of correct predictions (both positives and negatives) out of the total predictions. While accuracy is often used, it may not be the best metric in cases of class imbalance like predicting Networth_Next_Year, where the majority class might dominate the results.

- **Precision:** It is the proportion of true positive predictions out of all positive predictions. Precision focuses on minimizing false positives (FP).

- **F1 Score:** It is the harmonic mean of precision and recall. It provides a balance between the two metrics, ensuring that both false positives and false negatives are taken into account.

Since the goal is to reduce false negatives (FN), ensuring that Networth_Next_Year with negative values is predicted as negative when it should be, maximizing recall is important.

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6.2. Logistic Regression Model

6.2.1. Logistic regression result

The logistic regression model obtained using the train data set is as follows:

Logit Regression Results

Dep. Variable:	default	No. Observations:	2979
Model:	Logit	Df Residuals:	2939
Method:	MLE	Df Model:	39
Date:	Sat, 15 Feb 2025	Pseudo R-squ.:	-0.2425
Time:	14:02:56	Log-Likelihood:	-1904.8
converged:	False	LL-Null:	-1532.9
Covariance Type:	nonrobust	LLR p-value:	1.000

	coef	std err	z	P>|z|	[0.025	0.975]
Total_assets	-0.2048	1.75e+13	-1.17e-14	1.000	-3.43e+13	3.43e+13
Net_worth	-0.3608	0.215	-1.677	0.094	-0.783	0.061
Total_income	-1.9413	0.230	-8.443	0.000	-2.392	-1.491
Total_expenses	-1.8962	0.289	-6.553	0.000	-2.463	-1.329
Profit_after_tax	-0.3270	0.155	-2.112	0.035	-0.631	-0.024
PBDITA	0.2272	0.137	1.657	0.097	-0.041	0.496
PBT	-0.0858	0.161	-0.535	0.593	-0.400	0.229
Cash_profit	-0.4914	0.134	-3.668	0.000	-0.754	-0.229
PBDITA_as_perc_of_total_income	0.0359	0.098	0.368	0.713	-0.156	0.227
PBT_as_perc_of_total_income	0.0293	0.134	0.219	0.827	-0.233	0.292
PAT_as_perc_of_total_income	0.0971	0.123	0.788	0.431	-0.144	0.339
Cash_profit_as_perc_of_total_income	0.0212	0.110	0.194	0.847	-0.194	0.236
PAT_as_perc_of_net_worth	0.0955	0.060	1.587	0.112	-0.022	0.213
Sales	4.4044	nan	nan	nan	nan	nan
Total_capital	-0.2355	0.080	-2.938	0.003	-0.393	-0.078
Reserves_and_funds	-0.0449	0.094	-0.478	0.632	-0.229	0.139
Borrowings	0.1742	0.086	2.029	0.042	0.006	0.342
Current_liabilities_&_provisions	0.0197	0.091	0.217	0.828	-0.158	0.197
Shareholders_funds	0.5484	0.219	2.509	0.012	0.120	0.977
Cumulative_retained_profits	-0.1528	0.079	-1.936	0.053	-0.308	0.002
Capital_employed	0.0952	0.175	0.543	0.587	-0.248	0.439
TOL_to_TNW	-0.1372	0.075	-1.819	0.069	-0.285	0.011
Total_term_liabilities__to__tangible_net_worth	0.0030	0.073	0.041	0.968	-0.140	0.146
Contingent_liabilities__to__Net_worth_perc	-0.0451	0.046	-0.987	0.324	-0.135	0.044
Net_fixed_assets	0.0101	0.082	0.123	0.902	-0.152	0.172
Current_assets	0.0341	0.117	0.292	0.770	-0.194	0.263
Net_working_capital	-0.0918	0.053	-1.718	0.086	-0.197	0.013
Quick_ratio_times	0.0090	0.068	0.133	0.894	-0.124	0.142
Current_ratio_times	0.0346	0.067	0.520	0.603	-0.096	0.165
Debt_to_equity_ratio_times	-0.0447	0.090	-0.496	0.620	-0.221	0.132
Cash_to_current_liabilities_times	-0.2300	0.062	-3.729	0.000	-0.351	-0.109
Cash_to_average_cost_of_sales_per_day	0.1497	0.060	2.491	0.013	0.032	0.267
Creditors_turnover	-0.0623	0.052	-1.188	0.235	-0.165	0.041
Debtors_turnover	0.0354	0.052	0.680	0.496	-0.067	0.137
Finished_goods_turnover	-0.0745	0.058	-1.291	0.197	-0.188	0.039
WIP_turnover	0.1089	0.058	1.881	0.060	-0.005	0.222
Raw_material_turnover	-0.0426	0.047	-0.900	0.368	-0.135	0.050
Shares_outstanding	0.2989	0.081	3.695	0.000	0.140	0.457
EPS	0.1864	0.086	2.156	0.031	0.017	0.356
Adjusted_EPS	-0.2477	0.081	-3.070	0.002	-0.406	-0.090
Total_liabilities	-0.2048	1.75e+13	-1.17e-14	1.000	-3.43e+13	3.43e+13

Table 12. Logistic regression result-1

OBSERVATIONS:

- After addressing missing values and outliers, 41 variables were considered for logistic regression. However, many of these variables have very high p-values.
- The p-value of a variable helps determine its significance. If we set the significance level at 0.05 (5%), any variable with a p-value less than 0.05 is considered significant.
- However, the presence of multicollinearity among these variables can distort the p-values and affect the reliability of the model. Therefore, it is necessary to address multicollinearity before drawing conclusions from the results.

6.2.2. Model Performance Improvement

6.2.2.1. Test multicollinearity by checking variance inflation factor (VIF)

The VIF is a measure used to detect multicollinearity in a regression model. It quantifies how much the variance of a regression coefficient is inflated due to the correlation with other independent variables in the model.

```
Series before feature selection:

Total_assets                                  inf
Net_worth                                   36.11
Total_income                                95.04
Total_expenses                              56.89
Profit_after_tax                            20.59
PBDITA                                      12.04
PBT                                         21.69
Cash_profit                                 15.01
PBDITA_as_perc_of_total_income               6.34
PBT_as_perc_of_total_income                 12.87
PAT_as_perc_of_total_income                 11.05
Cash_profit_as_perc_of_total_income          7.85
PAT_as_perc_of_net_worth                     2.43
Sales                                      124.03
Total_capital                                3.96
Reserves_and_funds                           6.59
Borrowings                                   5.11
Current_liabilities_&_provisions             6.01
Shareholders_funds                          35.99
Cumulative_retained_profits                  4.45
Capital_employed                            23.50
TOL_to_TNW                                   3.89
Total_term_liabilities__to__tangible_net_worth   3.63
Contingent_liabilities__to__Net_worth_perc   1.31
Net_fixed_assets                             4.88
Current_assets                               9.68
Net_working_capital                          1.78
Quick_ratio_times                            2.93
Current_ratio_times                          2.91
Debt_to_equity_ratio_times                   5.58
Cash_to_current_liabilities_times            2.51
Cash_to_average_cost_of_sales_per_day        2.35
Creditors_turnover                           1.72
Debtors_turnover                             1.73
Finished_goods_turnover                      1.81
WIP_turnover                                 2.01
Raw_material_turnover                        1.43
Shares_outstanding                           3.42
EPS                                          4.76
Adjusted_EPS                                 4.13
Total_liabilities                             inf
dtype: float64
```

Table 13. VIF result – Before Dropping Variables

The VIF values which are higher than 5 suggests multicollinearity and hence have to be dropped.

The process involves first identifying and removing the variable with the highest VIF value. After each removal, the VIF values of the remaining variables are recalculated. This process is repeated by continually dropping the variable with the highest VIF value until all the remaining variables have a VIF value below 5.

Below is the result of dropping the values with VIF>5.

Series before feature selection:

Total_expenses	5.42
PBT	4.42
PBDITA_as_perc_of_total_income	2.85
PAT_as_perc_of_total_income	4.03
PAT_as_perc_of_net_worth	2.33
Total_capital	3.82
Reserves_and_funds	4.90
Borrowings	3.67
Current_liabilities_&_provisions	4.14
Cumulative_retained_profits	4.36
TOL_to_TNW	3.76
Total_term_liabilities__to__tangible_net_worth	3.53
Contingent_liabilities__to__Net_worth_perc	1.29
Net_fixed_assets	3.89
Net_working_capital	1.73
Quick_ratio_times	2.91
Current_ratio_times	2.87
Debt_to_equity_ratio_times	5.47
Cash_to_current_liabilities_times	2.47
Cash_to_average_cost_of_sales_per_day	2.32
Creditors_turnover	1.71
Debtors_turnover	1.69
Finished_goods_turnover	1.80
WIP_turnover	1.95
Raw_material_turnover	1.41
Shares_outstanding	3.33
EPS	4.64
Adjusted_EPS	4.06

dtype: float64

Table 14. VIF result – After Dropping Variables

The logistic regression model is applied to the above data after removal of multicollinearity to generate the following results.

Logit Regression Results

Dep. Variable:	default	No. Observations:	2979
Model:	Logit	Df Residuals:	2952
Method:	MLE	Df Model:	26
Date:	Sat, 15 Feb 2025	Pseudo R-squ.:	-0.2845
Time:	14:03:17	Log-Likelihood:	-1969.0
converged:	True	LL-Null:	-1532.9
Covariance Type:	nonrobust	LLR p-value:	1.000

	coef	std err	z	P>|z|	[0.025	0.975]
PBT	-0.4149	0.067	-6.227	0.000	-0.545	-0.284
PBDITA_as_perc_of_total_income	-0.0078	0.063	-0.124	0.901	-0.131	0.115
PAT_as_perc_of_total_income	0.1131	0.073	1.550	0.121	-0.030	0.256
PAT_as_perc_of_net_worth	0.1247	0.058	2.159	0.031	0.011	0.238
Total_capital	-0.2751	0.076	-3.617	0.000	-0.424	-0.126
Reserves_and_funds	-0.0237	0.077	-0.308	0.758	-0.175	0.127
Borrowings	0.1718	0.067	2.554	0.011	0.040	0.304
Current_liabilities_&_provisions	0.1063	0.063	1.682	0.093	-0.018	0.230
Cumulative_retained_profits	-0.1233	0.075	-1.652	0.099	-0.270	0.023
TOL_to_TNW	-0.1479	0.072	-2.045	0.041	-0.290	-0.006
Total_term_liabilities__to__tangible_net_worth	-0.0419	0.070	-0.600	0.549	-0.179	0.095
Contingent_liabilities__to__Net_worth_perc	-0.0296	0.044	-0.672	0.502	-0.116	0.057
Net_fixed_assets	-0.0454	0.070	-0.651	0.515	-0.182	0.091
Net_working_capital	-0.0555	0.050	-1.101	0.271	-0.154	0.043
Quick_ratio_times	0.0254	0.066	0.383	0.702	-0.105	0.155
Current_ratio_times	0.0260	0.065	0.399	0.690	-0.102	0.154
Debt_to_equity_ratio_times	-0.0049	0.087	-0.056	0.955	-0.176	0.166
Cash_to_current_liabilities_times	-0.2224	0.059	-3.764	0.000	-0.338	-0.107
Cash_to_average_cost_of_sales_per_day	0.1161	0.058	1.997	0.046	0.002	0.230
Creditors_turnover	-0.0319	0.051	-0.624	0.533	-0.132	0.068
Debtors_turnover	0.0690	0.050	1.367	0.172	-0.030	0.168
Finished_goods_turnover	-0.0427	0.056	-0.760	0.447	-0.153	0.067
WIP_turnover	0.1060	0.056	1.895	0.058	-0.004	0.216
Raw_material_turnover	-0.0088	0.046	-0.192	0.848	-0.099	0.081
Shares_outstanding	0.3658	0.077	4.738	0.000	0.215	0.517
EPS	0.1500	0.082	1.833	0.067	-0.010	0.310
Adjusted_EPS	-0.2223	0.077	-2.888	0.004	-0.373	-0.071

Table 15. Logistic regression result-2

The model includes several insignificant features with p-values greater than 0.05. Therefore, we sequentially remove variables with p-values exceeding 0.05 and re-fit the model to assess the p-values. This process is repeated until only the significant variables, which contribute meaningfully to the model, remain.

Below is the logistic regression model after removing the insignificant variables from the data set.

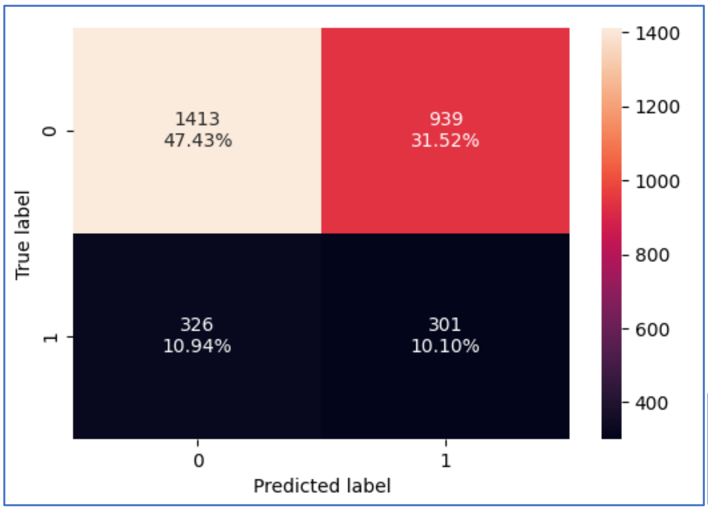
```
                           Logit Regression Results
==============================================================================
Dep. Variable:                default   No. Observations:                 2979
Model:                          Logit   Df Residuals:                     2969
Method:                           MLE   Df Model:                            9
Date:                Sat, 15 Feb 2025   Pseudo R-squ.:                 -0.2907
Time:                        14:03:22   Log-Likelihood:                -1978.5
converged:                       True   LL-Null:                       -1532.9
Covariance Type:            nonrobust   LLR p-value:                     1.000
===========================================================================================================
                                          coef    std err          z      P>|z|      [0.025      0.975]
-----------------------------------------------------------------------------------------------------------
PBT                                    -0.3452      0.058     -5.940      0.000      -0.459      -0.231
PAT_as_perc_of_net_worth                0.1993      0.049      4.066      0.000       0.103       0.295
Total_capital                          -0.2657      0.074     -3.594      0.000      -0.411      -0.121
Borrowings                              0.1225      0.053      2.332      0.020       0.020       0.225
Cumulative_retained_profits            -0.1115      0.053     -2.108      0.035      -0.215      -0.008
TOL_to_TNW                             -0.1668      0.043     -3.921      0.000      -0.250      -0.083
Cash_to_current_liabilities_times      -0.2075      0.054     -3.827      0.000      -0.314      -0.101
Cash_to_average_cost_of_sales_per_day   0.1295      0.055      2.362      0.018       0.022       0.237
Shares_outstanding                      0.3664      0.077      4.778      0.000       0.216       0.517
Adjusted_EPS                           -0.0989      0.050     -1.986      0.047      -0.197      -0.001
===========================================================================================================
```

Table 16. Logistic regression result-3

OBSERVATIONS:

- As all the p values are lesser than 0.05, we can consider features in **X_train_31** comprising **10 variables** as the final ones and **lg** as the final model.

6.2.2.3. Checking model performance on the training set



Training performance:

	Accuracy	Recall	Precision	F1
0	0.58	0.48	0.24	0.32

Fig 27. Confusion matrix-Train set

OBSERVATIONS:

- The false negatives constituted to around 10.94%.
- The model has a moderate accuracy of 0.58 and relatively moderate recall of 0.48.

6.2.2.5. ROC-AUC

The ROC-AUC curve is a performance measurement tool for binary classification models. It helps to evaluate how well a model distinguishes between classes, with a focus on the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) across different threshold values.

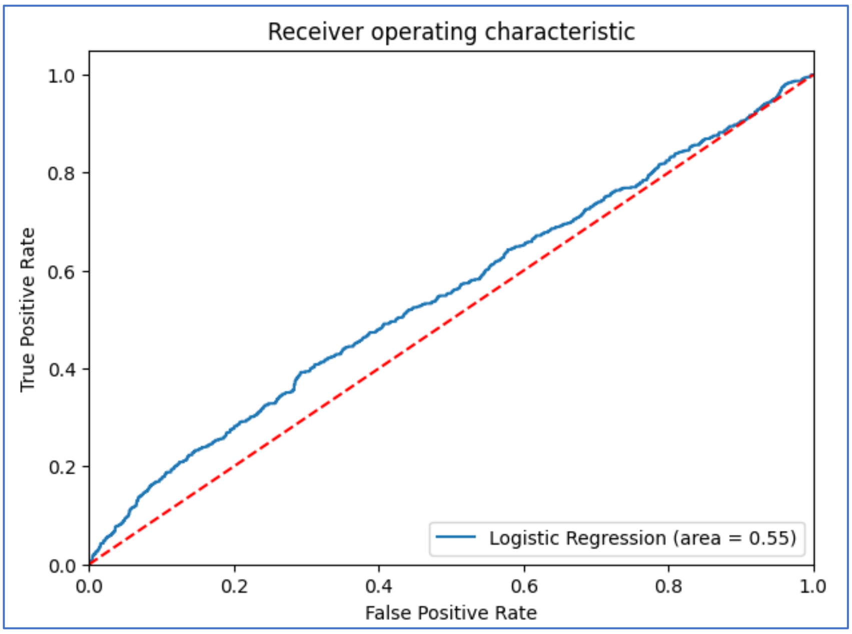


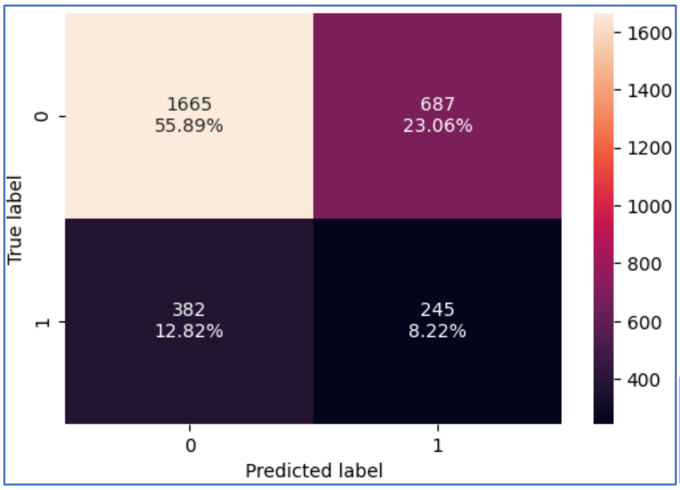
Fig 28. ROC-AUC-Train set

OBSERVATIONS:

- The model is performing satisfactory on training set.

6.2.2.7. Optimal threshold using ROC-AUC curve

The optimum threshold using the ROC-AUV curve is **0.5232.**



Training performance:

	Accuracy	Recall	Precision	F1
0	0.64	0.39	0.26	0.31

Fig 29. Confusion matrix-Train set Optimum threshold

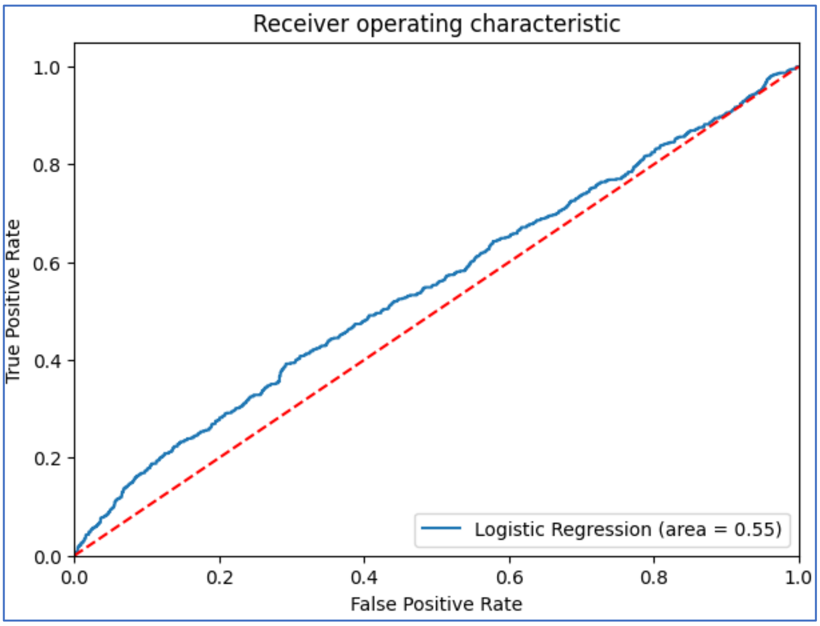


Fig 30. ROC-AUC-Train set Optimum threshold

OBSERVATIONS:

- The false negatives have increased from 10% to 12%.
- The accuracy has increased however the recall of model has reduced.
- The model is giving a poor performance.

6.2.2.8. Precision-Recall Curve

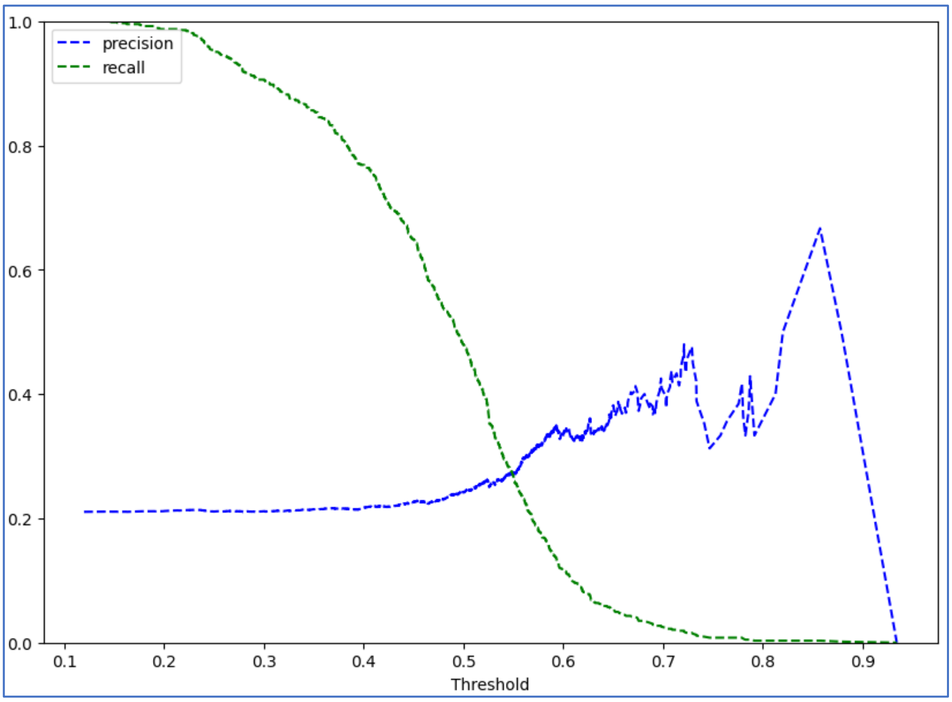
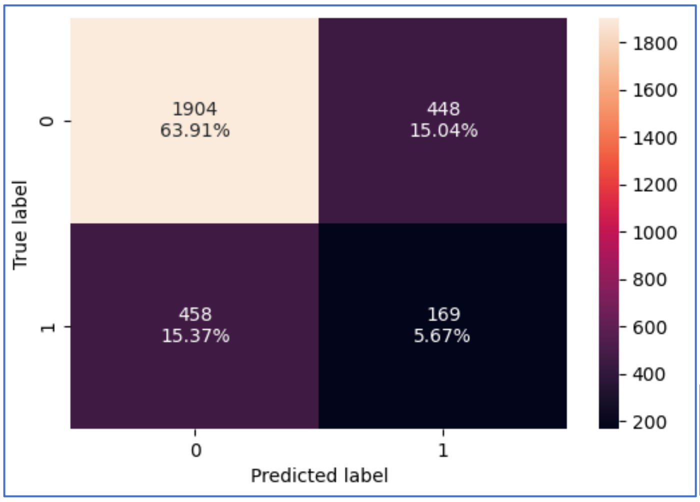


Fig 31. Precision-Recall curve

OBSERVATIONS:

- At the threshold of 0.55, we get balanced recall, but the precision is found to be imbalanced.



Training performance:

	Accuracy	Recall	Precision	F1
0	0.70	0.27	0.27	0.27

Fig 32. Confusion matrix-Train set Threshold

OBSERVATIONS:

- The recall has reduced drastically.
- There is no improvement in the model performance.

6.2.2.9. Training performance comparison

Training performance comparison:

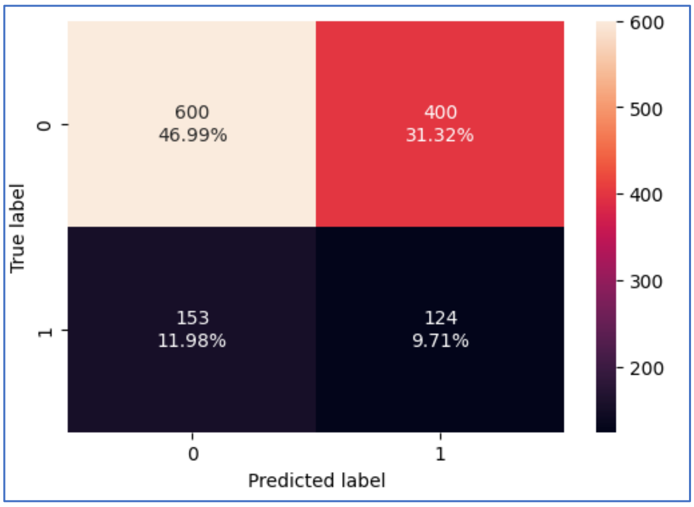
	Logistic Regression statsmodel	Logistic Regression-0.52 Threshold	Logistic Regression-0.55 Threshold
Accuracy	0.58	0.64	0.70
Recall	0.48	0.39	0.27
Precision	0.24	0.26	0.27
F1	0.32	0.31	0.27

Table 17. Training performance comparison

OBSERVATIONS:

- The recall is higher (0.48), with logistic regression stats model.
- The optimum threshold using the ROC-AUC decreases the recall drastically.
- However, the accuracy increases with optimum threshold, while precision and F1 being almost constant.

6.2.2.10. Checking model performance on the test set



Test performance:

	Accuracy	Recall	Precision	F1
0	0.57	0.45	0.24	0.31

Fig 33. Confusion matrix-Test set

OBSERVATIONS:

- The model has a moderate recall of 0.45 and relatively low recall precision of 0.24.
- The false negatives account to about 11.98%.

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6.2.2.11. ROC-AUC

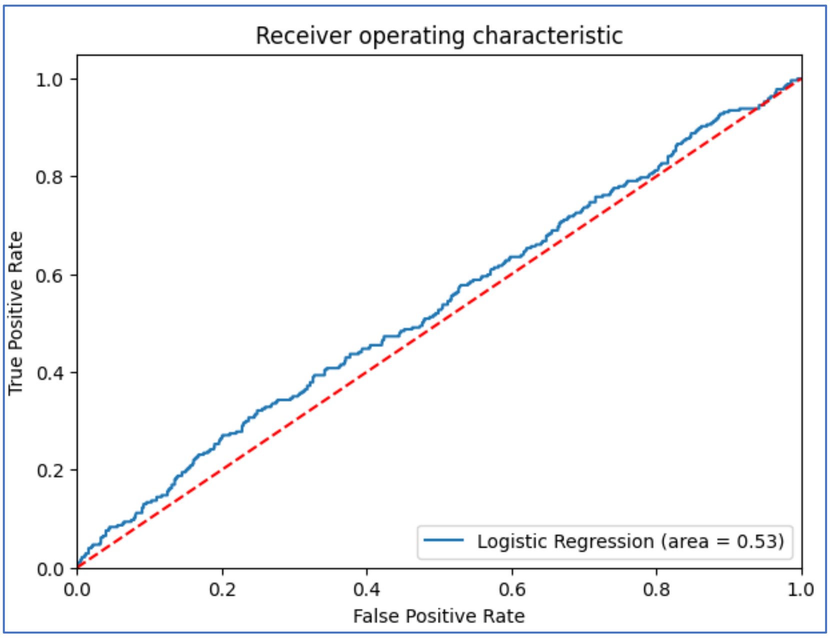
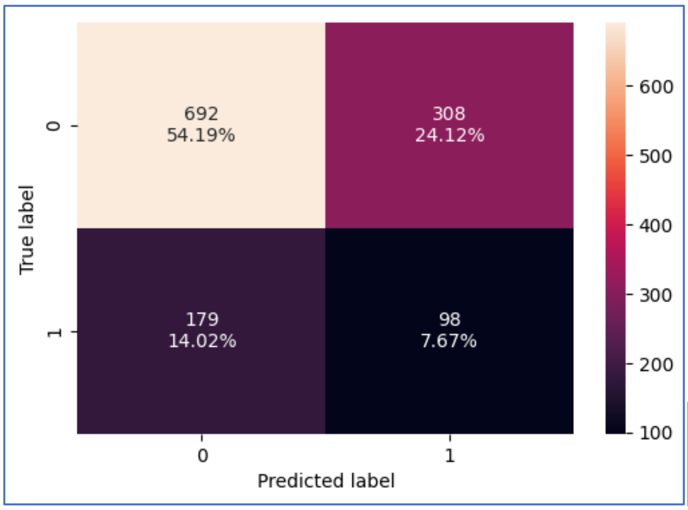


Fig 34. ROC-AUC-Test set

OBSERVATIONS:

- The model is performing not so good on test set.

6.2.2.12. Optimal threshold using ROC-AUC curve



Test performance:

	Accuracy	Recall	Precision	F1
0	0.62	0.35	0.24	0.29

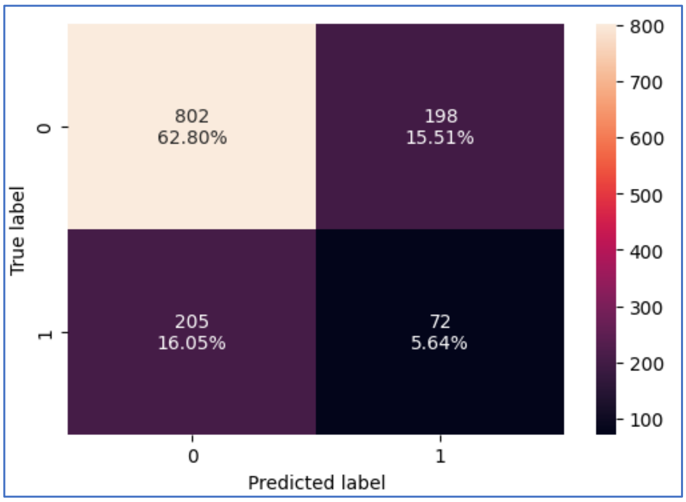
Fig 35. Confusion matrix-Test set Optimum threshold

OBSERVATIONS:

- The recall has dropped from 0.45 to 0.35.

6.2.2.13. Precision-Recall Curve

At the threshold of 0.55, we get the below matrix.



Test performance:

	Accuracy	Recall	Precision	F1
0	0.68	0.26	0.27	0.26

Fig 36. Confusion matrix-Test set Threshold

OBSERVATIONS:

- The recall has dropped to 0.26.
- There's no improvement in the model performance.

6.2.2.14. Test performance comparison

Test set performance comparison:

	Logistic Regression statsmodel	Logistic Regression-0.52 Threshold	Logistic Regression-0.55 Threshold
Accuracy	0.57	0.62	0.68
Recall	0.45	0.35	0.26
Precision	0.24	0.24	0.27
F1	0.31	0.29	0.26

Table 18. Test performance comparison

OBSERVATIONS:

- The recall is moderate (0.45), with logistic regression stats model.
- The optimum threshold doesn't seem to increase the recall value, and decreases recall to 0.35 and 0.26.

6.2.4. Logistic Regression Model with Balanced Data

Synthetic Minority Over-sampling Technique (SMOTE) is a popular technique used for handling class imbalance in classification problems. When dealing with imbalanced datasets, the model tends to predict the majority class more often and may fail to identify the minority class effectively, leading to poor recall.

Below is the classification report for Train set using Logistic Regression model

Classification Report for Train Set:

	precision	recall	f1-score	support
0.0	0.56	0.51	0.54	2352
1.0	0.55	0.60	0.58	2352
accuracy			0.56	4704
macro avg	0.56	0.56	0.56	4704
weighted avg	0.56	0.56	0.56	4704

Table 19. Logistic Regression (Balanced) Classification report - Train set

Below is the classification report for Test set using Logistic Regression model.

Classification Report for Test Set:

	precision	recall	f1-score	support
0.0	0.81	0.48	0.60	1000
1.0	0.24	0.59	0.34	277
accuracy			0.51	1277
macro avg	0.52	0.54	0.47	1277
weighted avg	0.69	0.51	0.55	1277

Table 20. Logistic Regression (Balanced) Classification report - Test set

OBSERVATIONS:

- The recall score is 0.60 for train set and 0.59 for test set. The model is moderately good at detecting positive cases, but there’s room for improvement
- The precision is 0.55 and 0.24 in the above model. The higher precision on the test set could imply the model is more confident in its positive predictions when applied to new data, but the relatively low precision on the training set suggests there’s still some noise or incorrect predictions.

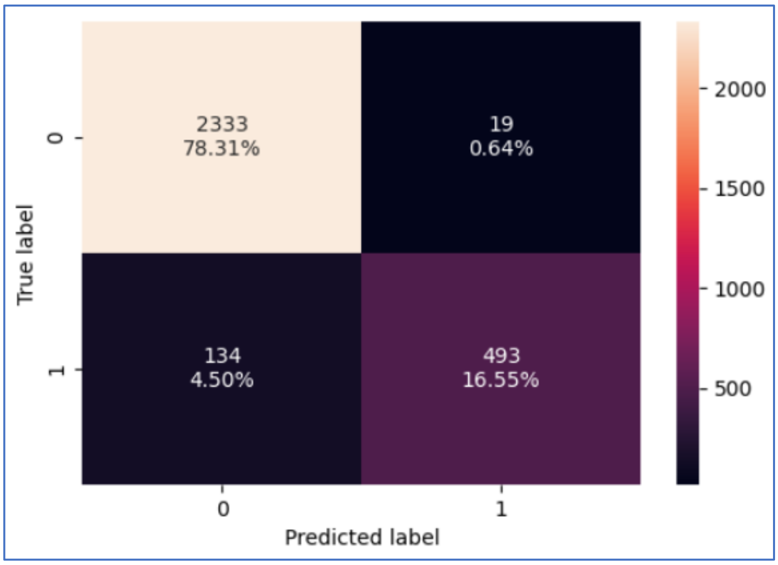
6.2.6. Insights of Logistic Regression Model

- The logistic regression model gives a generalized performance on training and test set.
- The highest recall is 0.48 and 0.45 on both training and test set respectively.
- Using the model with 0.55 threshold does not give high recall but gives higher accuracy and stable precision and F1 scores in both the train and test set.
- The Logistic regression stats model will help to identify potential defaulters.

6.3. Random Forest Model

Random Forests can perform well with imbalanced datasets. This is because the ensemble nature and the randomness in selection allow the model to focus on learning both classes without being biased towards the majority class.

Training performance:



Training performance

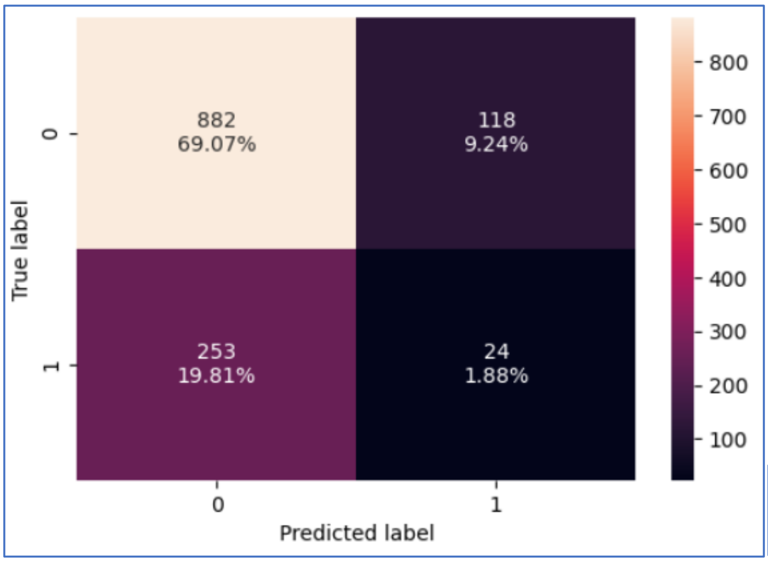
	Accuracy	Recall	Precision	F1
0	0.95	0.79	0.96	0.87

Fig 37. Random Forest – Train set

OBSERVATIONS:

- The model is having a high recall of 0.79, with 4.5% being the false negatives.
- The accuracy and precision also seem to be working well.

Test performance:



Testing performance

	Accuracy	Recall	Precision	F1
0	0.71	0.09	0.17	0.11

Fig 38. Random Forest – Test set

OBSERVATIONS:

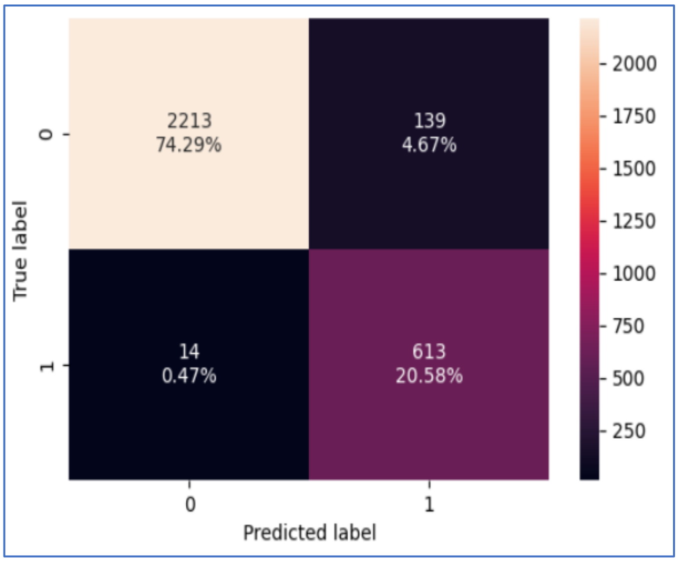
- The false negatives increase from 4.5% in the train set to 19.81 in the test set.
- The recall drops drastically to 0.09.
- The precision also is low.

6.3.2. Model Performance Improvement

6.3.2.1. Random Forest Model with class weights

The Random Forest model is obtained by using class weights set to 'balanced'.

Below is the performance of Train set.



Training performance

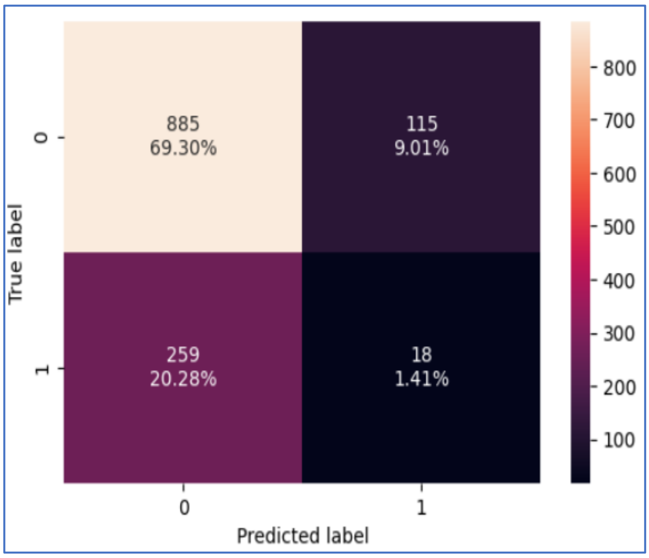
	Accuracy	Recall	Precision	F1
0	0.95	0.98	0.82	0.89

Fig 39. Random Forest (balanced) – Train set

OBSERVATIONS:

- The model is having a high recall of 0.98, with 0.47% being the false negatives.
- The accuracy and precision also seem to be working well.

Below is the performance of Test set.



Testing performance

	Accuracy	Recall	Precision	F1
0	0.71	0.06	0.14	0.09

Fig 40. Random Forest (balanced) – Test set

OBSERVATIONS:

- The false negatives increase from 0.47% in the train set to 20.28% in the test set.
- The recall drops drastically to 0.06.
- The precision also is low.

6.3.2.2. Hyperparameter Tuning using GridSearchCV

Hyperparameter tuning using GridSearchCV involves systematically testing a range of hyperparameters for a model to find the optimal combination that results in the best performance.

```
Fitting 5 folds for each of 72 candidates, totalling 360 fits
Best parameters found by GridSearchCV: {'bootstrap': True, 'max_depth': 10, 'max_features': 'sqrt', 'min_samples_leaf': 2, 'min_samples_split': 2, 'n_es
timators': 200}
```

Below is the performance of Train set.

Training performance

	Accuracy	Recall	Precision	F1
0	0.95	0.98	0.82	0.89

Table 21. Tuned Random Forest Classification report - Train set

OBSERVATIONS:

- The recall is about 0.98 which is high.
- The precision and F1 scores are high.

Below is the performance of Test set.

Test performance

	Accuracy	Recall	Precision	F1
0	0.71	0.06	0.14	0.09

Table 22. Tuned Random Forest Classification report - Test set

OBSERVATIONS:

- The recall is too low, about 0.06.
- The model does not perform well.

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6.3.4. Insights of Random Forest Model

- The random forest model, random forest model with balanced class weights and hyperparameter tuned random forest model is likely overfitting on the training data. It has learned to perform well on the training set but fails to generalize to new, unseen data.
- The increase in false negatives is consistent with the recall drop, and it suggests that the model is not generalizing well to the test data.
- The model might be too sensitive to the training data patterns and is unable to correctly predict the positive cases in the test data.
- Hence the random forest model is not a good fit.

6.4. Model Comparison

Training Performance:

	Logistic Regression (Stats)	Logistic Regression – 0.52	Logistic Regression – 0.52	Random Forest	Random Forest - Balanced	Random Forest - Tuned
Accuracy	**0.58**	0.64	0.70	0.95	0.95	0.95
Recall	**0.48**	0.39	0.27	0.79	0.98	0.98
Precision	**0.24**	0.26	0.27	0.96	0.82	0.82
F1	**0.32**	0.31	0.27	0.87	0.89	0.89

Table 23. Model comparison – Train set

Test Performance:

	Logistic Regression (Stats)	Logistic Regression – 0.52	Logistic Regression – 0.52	Random Forest	Random Forest - Balanced	Random Forest - Tuned
Accuracy	**0.57**	0.62	0.68	0.71	0.71	0.71
Recall	**0.45**	0.35	0.26	0.09	0.06	0.06
Precision	**0.24**	0.24	0.27	0.17	0.14	0.14
F1	**0.31**	0.29	0.26	0.11	0.09	0.09

Table 24. Model comparison – Test set

OBSERVATIONS:

- A total of six models were developed using the given dataset. However, the performance of both the Logistic Regression and Random Forest models was not satisfactory.
- Among the Logistic Regression models, the one built using the stats library outperformed the others, delivering a better recall rate of 0.48 on the training data and 0.45 on the test data.
- Based on this performance, the Logistic Regression model is considered the most suitable for this dataset.
- The Logistic Regression model created using the stats library yielded the highest recall scores, suggesting it is better at identifying the positive class compared to other models. This is especially useful in situations where minimizing false negatives is a priority.

6.5. Important Features

Below is the coefficient interpretation of the most significant variables in the data set.

	PBT	PAT_as_perc_of_net_worth	Total_capital	Borrowings	Cumulative_retained_profits	TOL_to_TNW	Cash_to_current_liabilities_times	Cash_to_average_cost_of_sales_per_day	Shares_outstanding	Adjusted_EPS
Odds	0.71	1.22	0.77	1.13	0.89	0.85	0.81	1.14	1.44	0.91
Change_odd%	-29.19	22.06	-23.33	13.03	-10.55	-15.36	-18.74	13.82	44.25	-9.42

Table 25. Important Feature - Coefficient Interpretations

OBSERVATIONS:

- **PBT (Profit Before Tax)**: A decrease in PBT lowers the likelihood of the positive class, with a 29.19% drop in odds, indicating a strong inverse relationship with the target.
- **PAT_as_perc_of_net_worth**: A higher PAT relative to net worth increases the likelihood of the positive class, with a 22.06% increase in odds.
- **Total_capital**: An increase in total capital decreases the likelihood of the positive class, with a 23.33% drop in odds.
- **Borrowings**: Higher borrowings increase the likelihood of the positive class, with a 13.03% rise in odds, suggesting financial stress.
- **Cumulative_retained_profits**: Lower retained profits reduce the likelihood of the positive class, with a 10.55% decrease in odds, indicating potential instability.**TOL_to_TNW**: A higher TOL to TNW ratio decreases the likelihood of the positive class, with a 15.36% drop in odds, i
- ndicating higher leverage risk.
- **Cash_to_current_liabilities_times**: A lower cash-to-current liabilities ratio decreases the likelihood of the positive class, with an 18.74% drop in odds, suggesting liquidity issues.
- **Cash_to_average_cost_of_sales_per_day**: A higher ratio increases the likelihood of the positive class, showing better liquidity relative to sales costs.
- **Shares_outstanding**: A higher number of shares outstanding significantly increases the likelihood of the positive class, with a 44.25% rise in odds, indicating lower risk.
- **Adjusted_EPS**: Lower Adjusted EPS decreases the likelihood of the positive class, with a 9.42% drop in odds, indicating financial weakness.

7. Actionable Insights & Recommendations

7.1. Focus on Debt Management:

- Companies with high borrowings and high TOL to TNW ratio are more likely to face financial distress. Introduce debt reduction strategies, such as paying down high-interest loans or renegotiating terms to reduce financial pressure.
- Companies with a low Cash to Current Liabilities ratio are at risk of being unable to cover short-term obligations. Encourage businesses to focus on improving their liquidity positions by maintaining sufficient cash reserves.
- Strengthen cash management policies to ensure that sufficient liquidity is available for short-term obligations. Implementing automated cash flow forecasting tools can help prevent liquidity shortfalls.

7.2. Profitability and Financial Stability:

- The models indicate that declining Profit Before Tax (PBT) and PAT as a percentage of net worth are strong indicators of financial stress. Undertake a profitability audit and identify underperforming areas that can be optimized. Encourage diversification into more profitable markets or products to reduce dependency on a single revenue source.

7.3. Credit Risk Management:

- Investors should use models that prioritize recall as reducing false negatives is critical. Implement a tool based on the model's predictions to evaluate potential investment opportunities.
- Use advanced credit risk assessment tools that integrate machine learning algorithms (like logistic regression) to predict default probabilities. Investors should rely on these tools to analyze historical data and make informed decisions about which companies to invest in or lend to.

7.4. Proactive Financial Risk Mitigation:

- A real-time financial distress alert system should be put in place to track the financial health of companies continuously. This system should use key financial ratios to trigger alerts when thresholds indicating financial risk are crossed.

- Implement an AI-driven early warning system that uses financial indicators to monitor and predict the likelihood of a company defaulting. Set thresholds for key indicators, and trigger alerts to allow businesses and investors to take timely corrective actions.

7.5. Debt Repayment and Refinancing Strategy:

- Initiate discussions with financial institutions for debt restructuring if a company shows signs of facing financial stress. This can help avoid a default scenario and ensure that debt payments are more manageable.

7.6. Strategic Partnerships and Equity Funding:

- Encourage companies to raise equity capital (through issuing shares) rather than increasing debt when in need of funds.
- Look for opportunities to raise equity capital through new investments, strategic partnerships, or public offerings.

7.7. Investor Education and Awareness:

- Investors should be educated about the key financial metrics that indicate risk. This can help them make informed decisions and reduce the likelihood of investing in distressed companies.

7.8. Long-term Financial Strategy:

- Encourage companies to adopt a holistic approach to their financial strategy by focusing on improving profitability, reducing debt, enhancing liquidity, and managing risk efficiently.